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An Experimental Study of Nonlinear Stiffness, Hysteresis, and Friction Effects in Robot Joints with Harmonic Drives and Torque Sensors

Abstract

Despite widespread industrial application of harmonic drives, the source of some elastokinetic phenomena and their impact on overall system behavior has not been fully addressed thus far. Some of these phenomena severely influence the behavior of robot arms, both in free and constrained motions, when the end effector is in contact with an environment. The primary goal of this study is to derive an effective, control-oriented model of a harmonic-drive-based robot joint. Systematic observations of an experimental robot with harmonic drives has revealed that the harmonic drive could not entirely transmit the input torque to the output shaft, due to a nonlinear meshing process between the flexible and circular spline teeth. The torque transmitted to the output shaft might saturate at a much lower value than expected (e.g., motor torque multiplied by the gear ratio). This phenomenon may severely influence the system behavior, particularly in force/impedance control tasks when full joint-torque capacity and wide bandwidth are needed. To understand the harmonic-drive behavior, as well as to derive a convenient form of its model, we have shown restrained motion experiments to be much more useful than free-motion experiments.

In this article, we also introduce mathematical models and describe experiments related to other physical phenomena, such as nonlinear stiffness, hysteresis, and soft windup. The goal of our modeling strategy was not to develop a precise and possibly complicated model, but to generate an appropriate model that could be easily used by control engineers to improve joint behavior. To visualize the developed model, equivalent mechanical and electrical schemes of the joint are introduced. Finally, a simple and reliable estimation procedure has been established for obtaining joint parameters, to ascertain the integrity of the proposed model.

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1. Introduction

Understanding and addressing the physical phenomena and inherent properties of robot drive systems is of paramount importance for achieving smooth low-speed motion and compliant behavior in contact situations. The basic components of robot drive systems are power amplifiers, actuators, transmission elements, and sensors. The transmission elements should embody important features such as zero backlash, low weight, low friction losses, compact size, and high torque capacity. These features strongly influence manipulator performance. One approach to circumvent these issues is to use high-torque or hydraulic actuators that do not require speed reduction between the motor and the link. However, high-torque/low-velocity motors are heavy and bulky, while hydraulic actuators require higher maintenance efforts than conventional electric motors. For this reason, it is important to study the dominant performance characteristics of robot transmissions (Schempf and Yoerger 1993). In this article, we focus on harmonic drive technology, which has received scant attention in robotics literature despite its outstanding properties, such as zero backlash and a high torque-to-weight ratio.

1.1. Historical Perspectives

Developed over 30 years ago, harmonic drives were primarily used in aerospace and military applications before becoming widely accepted for industrial applications. Over the last decade, they have been implemented extensively in robotics due to their unique and inherent properties, especially their virtual zero backlash.

As described in Dudley's well-known classic *Gear Handbook* (Dudley 1962), the harmonic drive concept represents "a radical departure from traditional mechanics

and requires entry into a new and challenging realm of *nonrigid mechanics*, or *elastokinesis*—the new field of elastic-body dynamics" (italics added for emphasis). This book provides a concise description of the device, a list of 10 unique properties, and a visionary set of application areas, including "servomechanisms" and "fine-coarse controllers," which are widely used in robotics today.

Despite its widespread industrial application, the harmonic drive has not attracted the expected attention of researchers, with the exception of those in the former Soviet Union (Margulis and Volkov 1987). Most of the available literature covers particular aspects of harmonic drives, such as kinematics and the geometry of tooth engagement (Oei 1991), kinematic error analysis (Ahmadian 1987), harmonic analysis of vibrations (Nye and Kraml 1991), etc. A detailed study of design aspects that includes the optimization of gear-teeth ratio and a profile to maximize surface contact is provided, to our knowledge, in only one publication (Oei 1991).

An attempt to explain the occurrence of periodic position errors can be found in Hsia (1988). The error was attributed to the geometry of the wave generator. A different approach, based on parametric curves to model flexspline deformation and reported in Ahmadian (1987), captures the first harmonic of the experimentally measured gear error. Nye and Kraml found that the first nine harmonics in gear error can be attributed to a lack of concentricity, misalignments, out-of-roundness, and tooth-placement errors (Nye and Kraml 1991). Hidaka et al. came to similar conclusions (Hidaka et al. 1990), but they also found that the amplitude of torsional vibrations increased as the load was increased. The experiments with the newer "S"-tooth profile drives (Nye and Kraml 1991; HD Systems 1992) demonstrated well-behaved gear error (about 50% of the previously obtained errors with the conventional tooth profile) that is free of higher harmonics. Apparently, the controversy about the source and genesis of harmonic drive errors has not been resolved thus far.

The transmission compliance mechanism is another controversial issue regarding the primary source of flexibility. Yanabe et al. studied the deformations of the flexspline under torque-loaded conditions (Yanabe et al. 1989) using finite-element stress analysis. A similar approach has been adopted by Hashimoto (Hashimoto 1989; Hashimoto, Kiyosawa, and Paul 1993). They examined strains on the flexspline from the point of view of torque sensing. They used 3D thick shells with 16 nodes in their finite-element stress analysis to find the best configuration of strain gauges for measuring the applied torque. Other researchers claim that the dominant source of elasticity is not the flexspline deformation, but the gear-tooth engagement zone, which determines the overall compliance of a harmonic drive as well as its torque capacity (Kiyosawa, Sasahara, and Ishikawa 1989). Finally, the wave-generator deformation is considered to be essential and dominant by Margulis and Volkov (Margulis and Volkov 1987). Clearly, a reliable finite-element analysis and experimental verification for all of a harmonic drive's components is still missing, making it impossible to highlight the contribution of each individual harmonic drive component to the overall transmission compliance.

Revealing the contributions to intuitive understanding and the analytical description of harmonic drive-based robot joints have been reported by Tuttle and Seering. They found that a harmonic drive can exhibit dynamic behavior surprisingly more turbulent than that of conventional gear transmissions (Tuttle 1992; Tuttle and Seering 1993). Their experimental observations show that the velocity response to step commands in motor current are contaminated not only by serious vibration, but also by unpredictable jumps. They explained that the unexpected state transition can occur when the transmission harnesses enough energy to push through a system resonance. The velocity time-response observations were used to guide the development of a series of increasingly complex models to describe the harmonic drive behavior. The most complex model takes into account frictional losses between the flex- and circular-spline teeth, as well as kinematic error constraints and nonlinear stiffness. The stiffness and friction were painstakingly adjusted to optimize the model performance.

Good, Sweet, and Strobel studied resonant behavior induced through dynamic interactions of motor drives and robot arms coupled by harmonic drives (Good, Sweet, and Strobel 1985). They found that the main sources of nonlinearity in the drives include the "hardening" stiffness characteristics of the harmonic drives, Coulomb friction, backlash, and current limiting in the velocity servos. This work demonstrates the inadequacy of conventional robot-arm control that is based on the widely accepted rigid-body model of the manipulator.

Similar results have been obtained by Karlen and coworkers (Karlen et al. 1990), with a dual-arm dexterous manipulator system that is based on harmonic drives and joint-torque sensors. They found that the harmonic drive is compliant and is nonlinear in compliance, and that it has a considerable amount of internal friction. They used velocity and torque feedback to improve joint behavior. Actuator-level torque feedback has also been proposed by Hogan et al. (Hogan, Rasolee, and Andary 1991), who studied the behavior of a robot arm during contact tasks and unrestrained motions, and the transitions between the two. The experimental results show that feedback loops cannot easily compensate for the internal nonlinearities of harmonic drives.

Experimental studies on harmonic drive behavior reveal numerous controversial issues regarding the source and

genesis of various phenomena that can negatively affect robot behavior, especially in hard contact tasks. This is the main reason why this research was deemed necessary.

1.2. Objectives

The preceding discussion would indicate that despite 30 years of history in harmonic drive technology, some physical phenomena have been discovered only recently. In this article, we describe an interesting phenomenon that we came across during experimentation with a newly designed multimanipulator experimental setup (Kircanski 1993; Hui et al. 1993). We found that a reduction gear may not necessarily act as a pure torque multiplier, but it may exhibit a "torque loss" after reaching a critical torque value—sometimes far below the maximum allowable torque of a selected unit. This effect appeared only in hard-contact tasks; in free motion, the torque was properly amplified, except for an amount due to frictional losses. The first objective of this article was to provide experimental evidence of this phenomenon, as well as a qualitative explanation of such behavior.

The second objective of this article was to find an optimal experimental strategy for obtaining reliable information on joint behavior. In contrast to the unrestrained motion experiments described in Tuttle (1992), most of our experiments considered the drive-system behavior in contact with a stiff environment. When the joint was unrestrained, we found that the intrinsic joint dynamics, including nonlinear stiffness, hysteresis, and some friction terms, were not influencing the system behavior because the operating point did not vary across the entire range of possible values. We found that the experimental facility joints did not exhibit the turbulent behavior described by Tuttle and Seering (Tuttle 1992; Tuttle and Seering 1993).

The third objective was to establish a simple mathematical representation of dominant physical effects that could be effectively utilized by controls engineers and mechanical designers of harmonic drive-based robot modules. First, a simple soft-windup model, in addition to the nonlinear stiffness model, was introduced. Also, hysteresis effects were studied and used to derive an adequate joint-model representation of moderate complexity. This model comprises two subsystems: a fast one with high-frequency modes reflecting wave-generator and rotor dynamics, and a slow one influenced by the output-link inertia properties. These two subsystems are coupled through the elastic deformation of the flexspline. Equivalent mechanical and electrical circuit representations of a harmonic drive-based joint were also introduced.

Finally, a systematic procedure based on a bias-compensating least-squares estimation algorithm was applied to capture the unknown parameters and to validate the assumed model structure. The estimation can be regarded as a fully automatic process that requires two experiments: one with the output link held fixed against a stiff obstacle, and the other that allows free motion of the link.

The article is organized as follows. Section 2 provides a short description of the experimental setup and control system. Section 3 describes torque-transmission characteristics of the harmonic drive gear, as well as the experimental results to rationalize the nonlinear stiffness behavior, soft-windup effect, and the Coulomb friction terms. Equivalent mechanical and electrical representations of a harmonic drive-based robot joint are introduced in Section 4. Section 5 presents a systematic procedure based on a bias-compensating least-squares estimation algorithm (Jia 1993). The article concludes with a brief discussion of the influence of various intrinsic joint parts on the overall drive-system behavior.

2. Experimental Setup

The literature overview provided in the previous section indicates that the research in the field of harmonic drive transmission phenomena must be supported with an appropriate experimental setup. This section will provide a short description of the experimental setup—the IRIS Facility—developed at the Robotics and Automation Laboratory of the University of Toronto (Hui et al. 1993). The facility is a versatile, reconfigurable and expandable setup composed of several robot arms that can be easily disassembled and reassembled to assume a multitude of configurations (Figure 1).

The constituent element of the system is the *joint module*, which has its own input and output links. Each module is equipped with a brushless DC motor that is coupled with a harmonic drive gear, and is instrumented with an optical encoder to measure the rotor angular displacement and a custom-designed torque sensor to measure the load torque about the joint axis. An external optical encoder can be mounted to measure the output-joint angle. The motor torque is indirectly measured through the motor current. The joint module was modified to allow direct measurement of the motor torque using high-accuracy tension/compression load cells.

Frameless motors are used to minimize the volume and mass of the joints. Every motor has three components: the armature assembly (stator), the field assembly (rotor), and a Hall device assembly. To minimize the volume and mass of the joint, the armature assembly and the rotor-shaft bearings are integrated into the heat-sinking shell. The joint module that was used for the experiments reported in this article (referred to as the "test-joint" in the text to follow) includes an RBE-0070 motor components set (Inland Motor Corp.) that is capable of delivering up to 0.38 N · m. The maximum continuous output power for

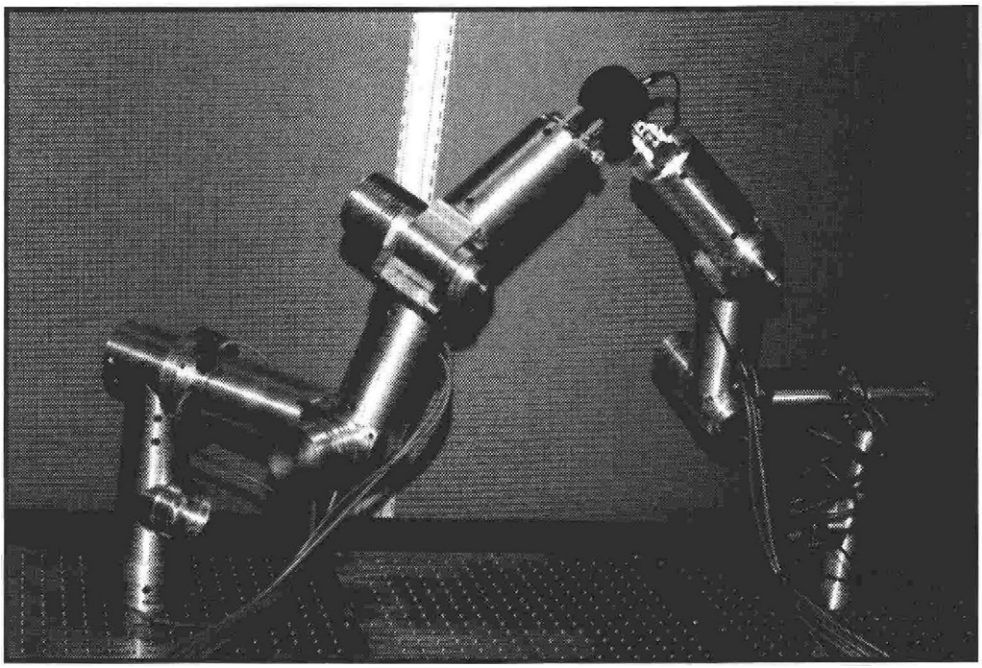


Fig. 1. IRIS arms.

this motor is 28 W. The motor is coupled to the compact, half-length cup, "S"-tooth profile, CSF-20-50-2A-GR gear components set (HD Systems Inc.) which provides a 50:1 speed reduction. The rated torque for this unit is 25 N · m, maximum average torque is 34 N · m, and momentary peak torque is 98 N · m.

The components of the harmonic drive (HD), wave generator (WG), flexspline (FS), and circular spline (CS), are placed concentrically around the rotor axis. The wave generator is a thin, raced ball-bearing assembly that is fitted into an elliptical plug. The flexspline is a thin cylinder cup having external teeth that are slightly smaller in pitch diameter than the circular spline. It is fitted over and elastically deflected by the wave generator. The circular spline is a rigid ring with internal teeth that engage the teeth of the flexspline across the major axis of the wave generator. The principle of operation can be easily explained from the cross-section of the harmonic drive shown in Figure 2. The flexspline is slightly smaller in diameter than the circular spline, usually resulting in its having two fewer teeth on its outer circumference (HD Systems 1990). It is held in an elliptical shape by the wave generator, and its teeth engage with the circular spline across the major axis of the ellipse. The engagement zone travels with the major elliptical axis as the wave generator rotates. Each full turn of the wave generator moves the flexspline two teeth in the opposite direction relative to the circular spline, provided that the circular spline is fixed. Otherwise, the circular spline will move two teeth in the direction of rotation relative to the flexspline. Thus, the speed of the wave generator is reduced approximately $N_t/2$ times, where N_t is the number of teeth on the circular spline (in our case $N_t = 100$).

A minimum-weight, custom-designed torque sensor is used for each IRIS joint to measure the load torque about the joint axis. The sensor has a stiffness coefficient at least 10 times higher than that of the harmonic drive itself. In contrast to the approach of Hashimoto (Hashimoto 1989), we have separated the torque sensing from the transmission to avoid the influences of flexspline vibration and nonlinear deformation on the torque signal. The torque-sensor electronics, including the amplifier and the signal conditioner, are placed close to the torque sensor inside the joint.

The test joint module was modified to allow direct measurement of the electromagnetic torque that was generated between the rotor and the stator. The armature assembly was augmented with two additional low-friction bearings, allowing the armature to rotate about the motor axis. The stator was rigidly coupled to an external load cell through a threaded circular rod (Figures 3 and 4). The load cell was fixed to the stationary frame to prevent the stator from rotating. In such configurations, the measured force is proportional to the reaction torque (the torque that is experienced by the armature assembly). This torque is equal to the electromagnetic torque that acts on the field assembly (or rotor). The main reasons for selecting such a measurement technique were: 1) relatively high accuracy (below 1% of rated output) of torque measurement is required, and 2) the relatively

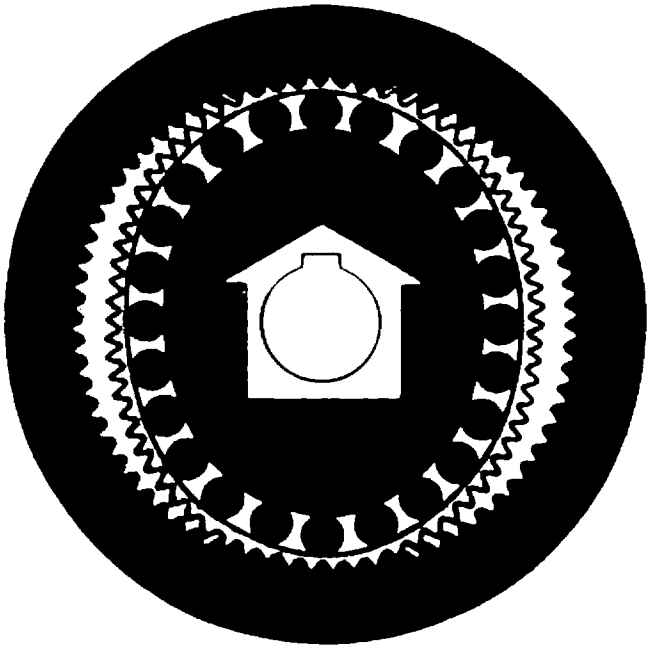


Fig. 2. Harmonic drive cross-section.

small torque (below 1 N · m) that can be produced by the motor assembly. We used the Transducer Techniques Inc. MLP-10 low-profile load cell, with a capacity of ± 10 lbs of tension/compression force measurement. This cell has exceptionally small nonlinearity (0.1% of rated output), hysteresis (0.1%), and nonrepeatability (0.05%). In addition to high accuracy, it has very high stiffness (0.7 MN/m), and a high natural frequency of 2,175 Hz. The stator assembly, including bearings, pins, and the load cell, was calibrated to allow reliable measurement of the motor torque.

The joint shown in Figure 3 has basically three ports: an input port (the A end), and two output ports (the C and F ends), with one of the latter being fixed to the input port. Depending on which output port is being fixed to the input one, the joint can represent either a pitch or roll type unit. By fixing the F end to the A end, the joint becomes a "pitch" module; fixing the C end to the A end makes it a "roll" joint.

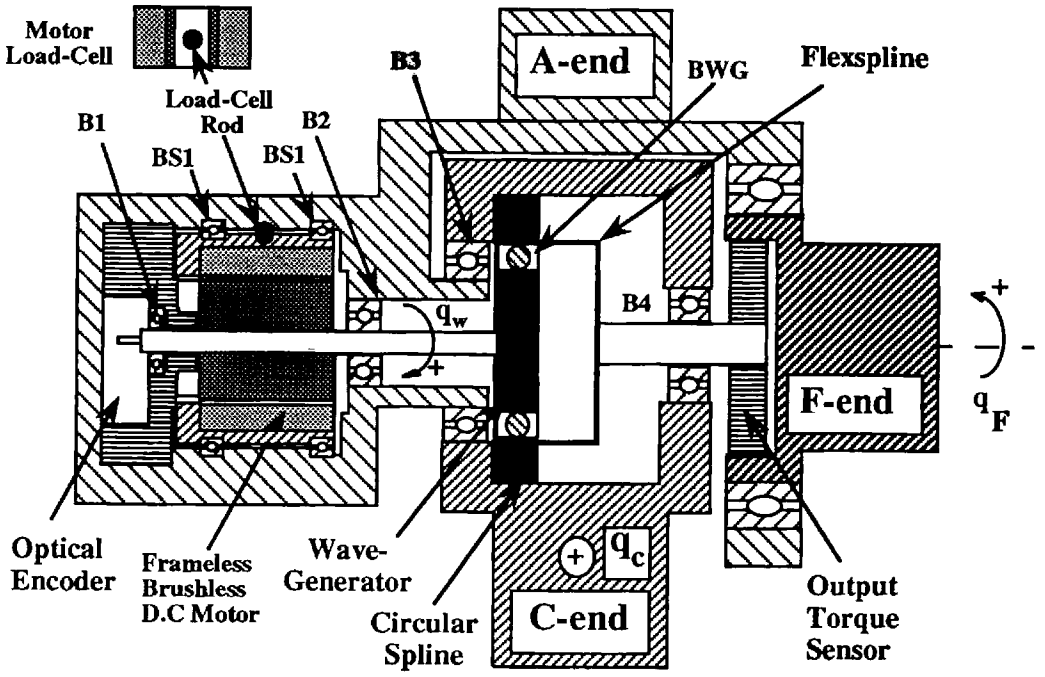
The angular coordinates of the wave generator, the F end and the C end are denoted by q_w, q_F, and q_c, respectively. Note that the wave-generator angle q_w is equal to the rotor angle q_r, assuming that the rotor shaft is perfectly rigid. By fixing one of the output ends to the input end, either q_F or q_c becomes zero. The wave-generator bearing (BWG) as well as other bearings within the joint ($Bi, i = 1, \ldots, 4$) are indicated in Figure 3 as well.

Controlled by a distributed computer system based on RISC processor nodes that are able to provide up to 80 MFLOPS per node (average speed is in the range of 10 to 20 MFLOPS), and a fast I/O system associated to each node allowing up to a 5 KHz sampling rate, the IRIS arms represent an ideal experimental environment for evaluating various estimation and control algorithms, including those described in this article (Kircanski 1993; Kircanski et al. 1994).

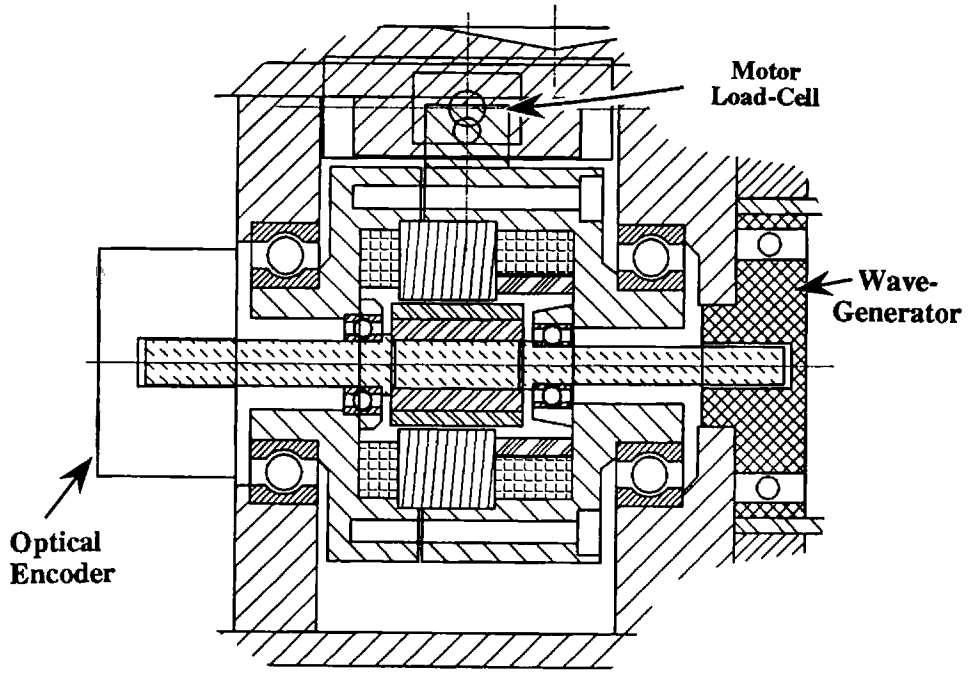
3. Experimental Results

The most promising way to highlight intrinsic joint phenomena and develop an appropriate modeling strategy, including an automated parameter-estimation procedure, is to conduct a series of experiments that incorporate both restrained and unrestrained joint motion. It has been demonstrated that experiments that assume the output link is in full contact with an environment of known stiffness are much more revealing compared with those allowing unrestrained motion of the link. A typical experimental environment is shown in Figure 4*A*. Notice that an external load cell with a known stiffness was used as a constraint, and a micrometer was used to precisely measure the angular displacement of the output link, instead of using a high-resolution external optical encoder (Figure 4*B*).

In this section, experimental results related to the torque-transmission characteristics in contact tasks, soft-windup effect, nonlinear stiffness, and friction phenomena are presented. The input/output torque-frequency response illustrates characteristic resonant behavior of a joint when the output link is in contact with a stiff environment.



a) Joint scheme



b) Details: Motor/wave-generator assembly

Fig. 3. Joint structure.

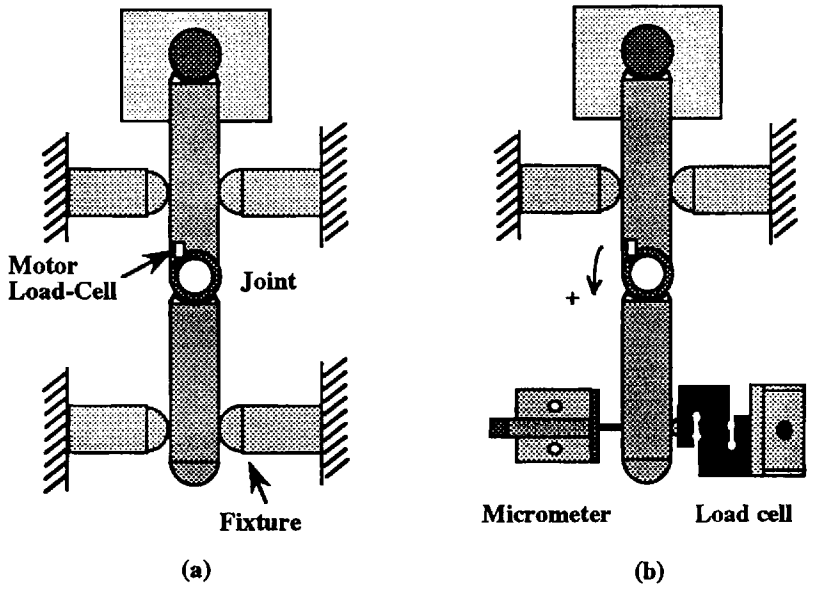


Fig. 4. Restrained motion setup.

As explained in Section 2, to separate the harmonic drive-related phenomena from the motor/amplifier phenomena we have installed a torque sensor assembly on the motor side (in addition to the torque sensor at the output port of the joint). The selected motor (RBE-00701) has a maximum torque ripple of 1% of the rated torque. This fluctuation includes the cogging torque and the ripple generated by the driver electronics (we used sinusoidal Inland's BLR series amplifier, which was designed to fit the characteristics of the selected motor).

One way to measure the motor torque is to use a small shaft-through torque sensor with slip rings for power and signal lines. The smallest "mini slip-ring torque sensor" that we could find on the market (A-Tech Inst. Ltd.) has the range ± 3.5 N · m, with 1% nonlinearity, 1% hysteresis, and nonrepeatability of 0.5% of the rated torque. Together, the nonlinearity, hysteresis, and nonrepeatability generate fluctuations which are much higher than that of the motor itself. For example, 1% nonlinearity on 3.5 N · m gives about 0.035 N · m, which is about 10% of the maximum motor torque for the selected motor. All effects together would deteriorate the measurement accuracy even more.

A custom-designed shaft-through torque sensor was the second choice, but we found that this approach would not solve the problems related to the increased inertia and friction due to the presence of such a device. In addition, the compliance of the sensor would influence the overall joint behavior, since the wave-generator angle would differ from the rotor angle due to the elastic deformation of the sensor itself. To correct this problem, an additional optical encoder would be needed on the wave-generator axis, and this would make the whole setup bulky and unreliable. For all of the abovementioned reasons, we installed additional low-friction bearings to allow the armature assembly to rotate about the motor axis (as shown in Figure 3). The stator was prevented from rotating by inserting a rigid rod through circular holes of larger diameter into the joint frame. The motor torque transmitted through the stator on the rod would rotate the rod, but that was obstructed by the use of the external load cell. The accuracy of the measurement achieved was in the range of up to 0.5% of the rated motor torque.

3.1. Torque-Transmission Characteristics

Let us assume that the links are in full contact with a stiff environment (Figure 4A). Assuming that a constant current is applied to the motor windings, the rotational velocity of the rotor, thus the back-emf, will be zero. The electromagnetic torque, measured by the input load cells, is opposed to the action of a constant force on the output link contact point. In this case, a reasonable assumption is that the output torque is proportional to the input torque, except for some small amount due to Coulomb friction inside the joint. In other words, the input-output torque characteristics form a straight line in the range $(-\tau_m, \tau_m)$, where τ_m is the maximum torque that can be delivered by the given DC motor and amplifier. Here, we suppose that the maximum stall torque of the motor is much lower that the rated torque of the gear. Nevertheless, our experiments proved that the input torque may not be fully passed onto the output link. The

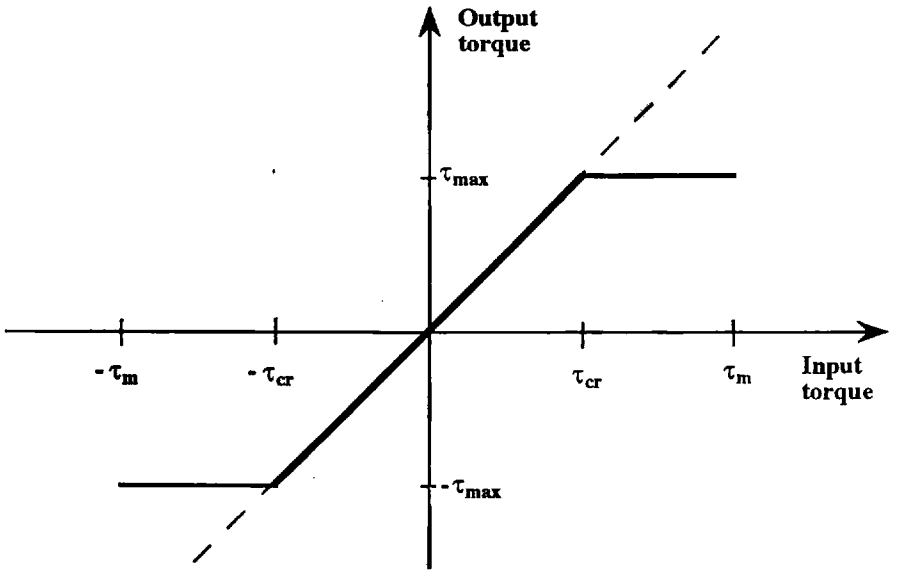


Fig. 5. Torque transmission characteristics.

actual input-output torque characteristics saturate in the region $\tau_{cr} \leq |\tau| \leq \tau_m$, where τ_{cr} stands for a "critical torque," which is in the range of 0 to τ_m, (Figure 5) in hard-contact tasks.

The existence of this phenomenon was experimentally verified by measuring the motor torque as well as the torque at the joint-torque sensor (Figure 6). In this experiment, the motor current is increased slowly toward the saturation of the amplifier. Figure 6 shows:

- measured motor torque scaled by the gear ratio, i.e., $N * \tau_m$,
- the output torque measured by the joint-torque sensor τ,
- the rotor angle q_r divided by the gear ratio, i.e., q_r/N, and
- the rotor angle scaled by the factor $-12.6/N \approx -0.25$,

where $N = 50$ is the gear ratio. The rotor angle is measured by the input optical encoder, with a resolution of $500 \cdot 4 = 2,000$ points per revolution.

Figure 6 shows that neither the joint-torque-sensor signal nor the rotor angle change smoothly: The stick-slip response is evident in both signals. They also reveal that the teeth-meshing process is nonuniform, as the smooth motion is from time to time interrupted by sudden rotor movements (see, for example, point A in Figure 6). The joint-torque sensor detects the jumps, and follows the rotor-angle evolution, truly reflecting the nonlinear meshing process in the harmonic drive. The motor-load cell signal does not show such explicit stick-slip response, since it is mainly influenced by the motor current value, which is, on the other hand, automatically regulated by the current-mode servo amplifier to match the commanded value (or ramp signal). The joint torque measured by the output-torque sensor saturates at about 5.3 N · m; far below the expected output torque of 12 N · m.

To capture the joint behavior under different excitations, we decreased the rise time 10 times compared to the previous case (Figure 7). Obviously, the time responses are very similar to the previous case. The major difference is in the smoothness of the curves, as the stick-slip effect is not visible and the response is smooth. Nevertheless, the output torque saturates (after an overshoot) at about 4.5 N · m; slightly lower than in Figure 6 (which is about 5.3 N · m). Thus, the critical torque depends on the rise time. After reaching the critical torque, both the input angle and the output torque are constant. This was also verified with an external load cell. The load-cell signal coincides with the joint-torque sensor signal (Kircanski 1993).

It is conjectured that the meshing process is a nonlinear phenomenon that prevents the rotor/wave-generator angle, as well as the joint torque, to reach predicted values: At a critical angle, the engaging teeth collide, preventing the wave generator from further rotation. The collision is probably the consequence of the deformation of the elliptically shaped flexspline cup. Its teeth are more and more twisted as the load increases. Actually, a small twist angle exists even at the no-load condition, providing virtually zero backlash of the gear. On the other hand, the meshing process is relatively smooth in unrestrained conditions. The measurements with the unrestrained output link have shown that the output torque is nearly propor-

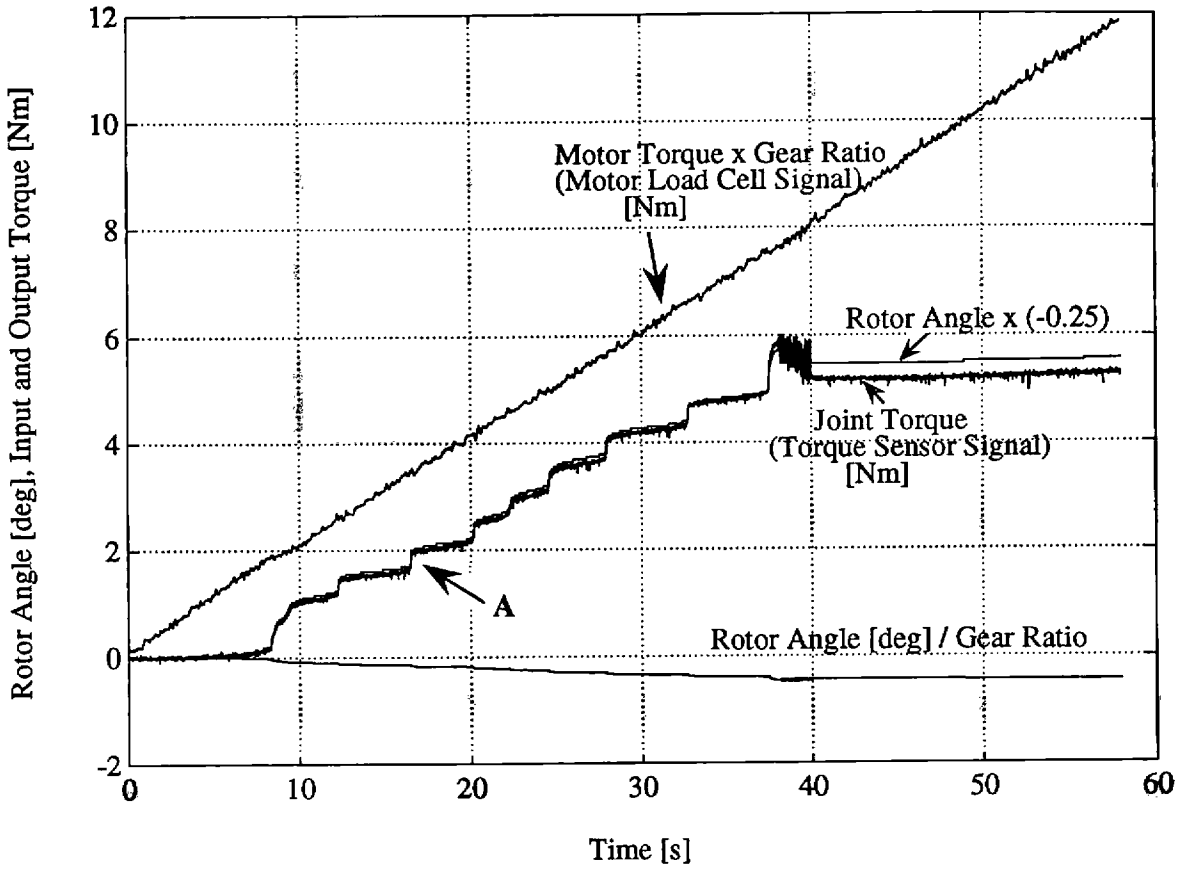


Fig. 6. Experimental evidence of torque saturation ("slow" ramp)—stiff surface contact.

tional to the input torque. Similarly, the motor torque is fully transmitted in "soft" contact tasks if the output angle changes along with a joint torque increase. Figures 8 and 9 illustrate the torque transmission in the case of a soft-contact task (we used a 5-mm sponge layer between the output link tip and a stiff surface). Since the joint angle was allowed to change its value, the rotor excursion was about five times higher than that in the case of a rigid contact. From the diagrams, we see that the output torque follows the input torque and saturates at an expected value.

3.2. *Soft Windup, Nonlinear Stiffness and Friction Characteristics*

There are three physical phenomena related to the behavior of the flexspline: nonlinear stiffness, quasi-backlash due to the soft-windup effect, and hysteresis (Good, Sweet, and Strobel 1985; Harmonic Drive 1990; HD Systems 1990; Tuttle 1992; HD Systems 1992). All these phenomena can be captured by the use of the experimental setup described above.

Suppose that the output link is fixed as shown in Figure 4*A*. By providing a sinusoidal input to the current-mode amplifier, the motor torque will change sinusoidally, provided that the amplifier is not exercised to its limits. In this experiment, the input signal frequency ω was held constant, while the bias current I was incremented from 0 to the saturation value. Thus, the commanded motor-current signal was $i_m = I + a\sin(\omega t)$, where the amplitude a was constant. We recorded the input and output torque signals, and the rotor angle q_w. Figure 10 shows three signals: 1) rotor angle q_w, 2) motor torque (measured by the motor load cell) scaled by the gear ratio $N = 50$, i.e., $N\tau_m$, and 3) the joint-torque sensor signal τ. Notice that the rotor angle follows the input signal, due to the rotational compliance of the joint. We can obtain the joint-stiffness coefficient for each mean value I. The stiffness coefficient is equal to the ratio of the amplitudes of the joint-torque sensor signal to amplitudes of the joint angular displacements. The joint-stiffness coefficient was obtained as a function of average applied torque. This is shown in Figure 11. We see that the torque-torsion characteristic has a parabolic shape while it is deformed toward

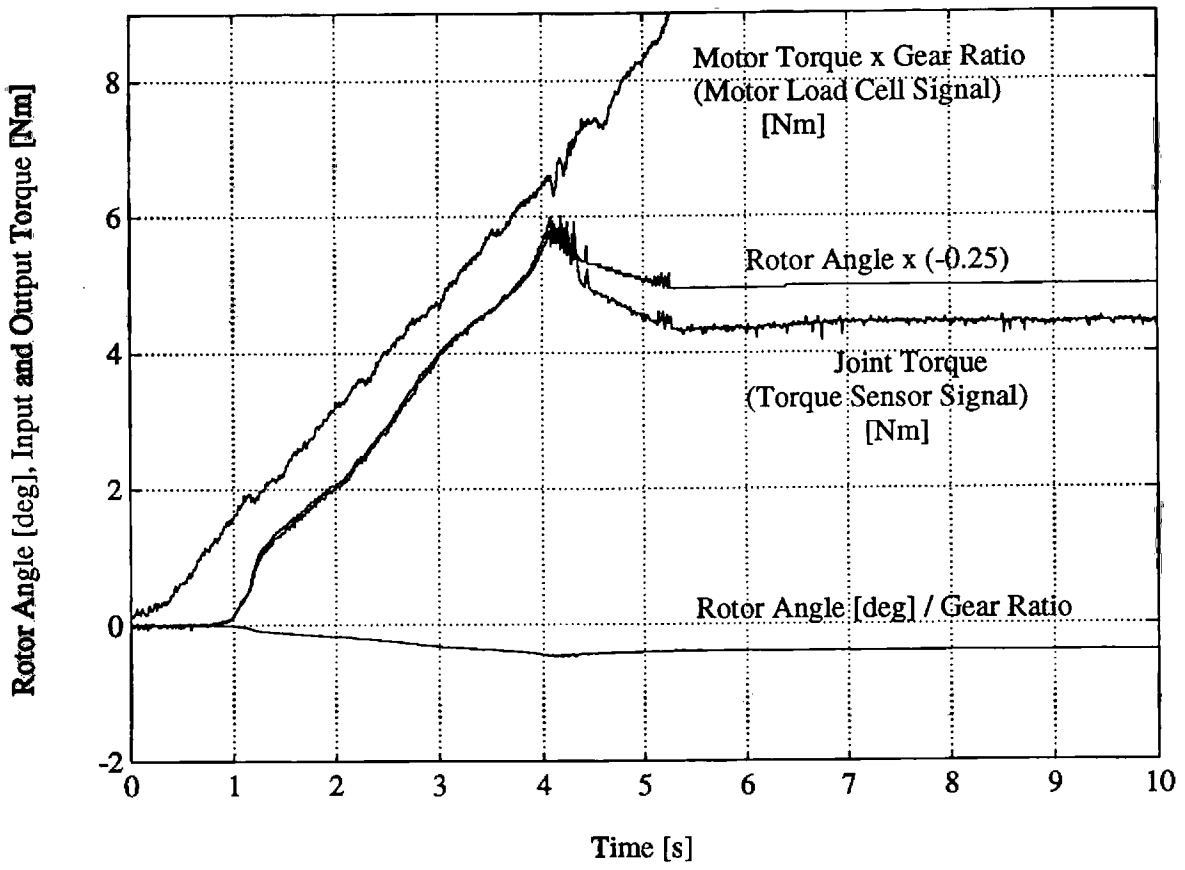


Fig. 7. Experimental evidence of torque saturation ("fast" ramp)—stiff surface contact.

the torque axis in the region from 0 to 1 N · m. In this region, the stiffness is lower due to the *soft-windup effect.*

We used the same experiment to observe the hysteresis effect due to Coulomb friction. We changed the motor current slowly to follow a sinusoidal function with zero mean value, but with different magnitudes (the amplitude was increased after every two cycles, while the frequency was held constant). As a consequence, the angular velocity of the rotor was gradually increased. We recorded the torque-sensor output as a function of angular displacement, and obtained the diagram shown in Figure 12. We see that when a certain motor torque is applied and then removed, the torsional displacement does not return to zero. This is the well-known hysteresis effect, which can be fully characterized by the diagram shown in Figure 12. The Coulomb friction term can be easily obtained from the hysteresis plot, since it is equal to the difference between the input motor torque and the torque-sensor output. Figure 13 shows the Coulomb friction torque as a function of torque-sensor output. As expected, the friction torque depends mostly upon the direction of rotation, and is nearly independent of velocity. Also, the hysteresis is shown as two values (or more) of i_m for each value of the torque output.

3.3. Frequency Response

The frequency response of a particular joint or of the entire robot arm at a given operating point (the robot configuration) is usually used to provide reliable information on the inertial properties in certain directions or about certain axes. With harmonic drive joints, such experiments can be dangerous, because the internal resonance can yield unpredictable jumps in velocity (Tuttle 1992; Tuttle and Seering 1993) and possibly damage the gear teeth. In resonance, large torque fluctuations can produce ratcheting and leave the harmonic drive in the "dedoidaled" condition, when the wave generator and the flexspline teeth are no longer centered in the circular spline, but instead are slipped over to one side (Nye and Kraml 1991). In this condition, the gear exhibits large variations in starting friction torque, while its lifetime is significantly reduced. We had to quit such experiments after a couple of trials, because the sinusoidal joint-torque signal was corrupted by extremely

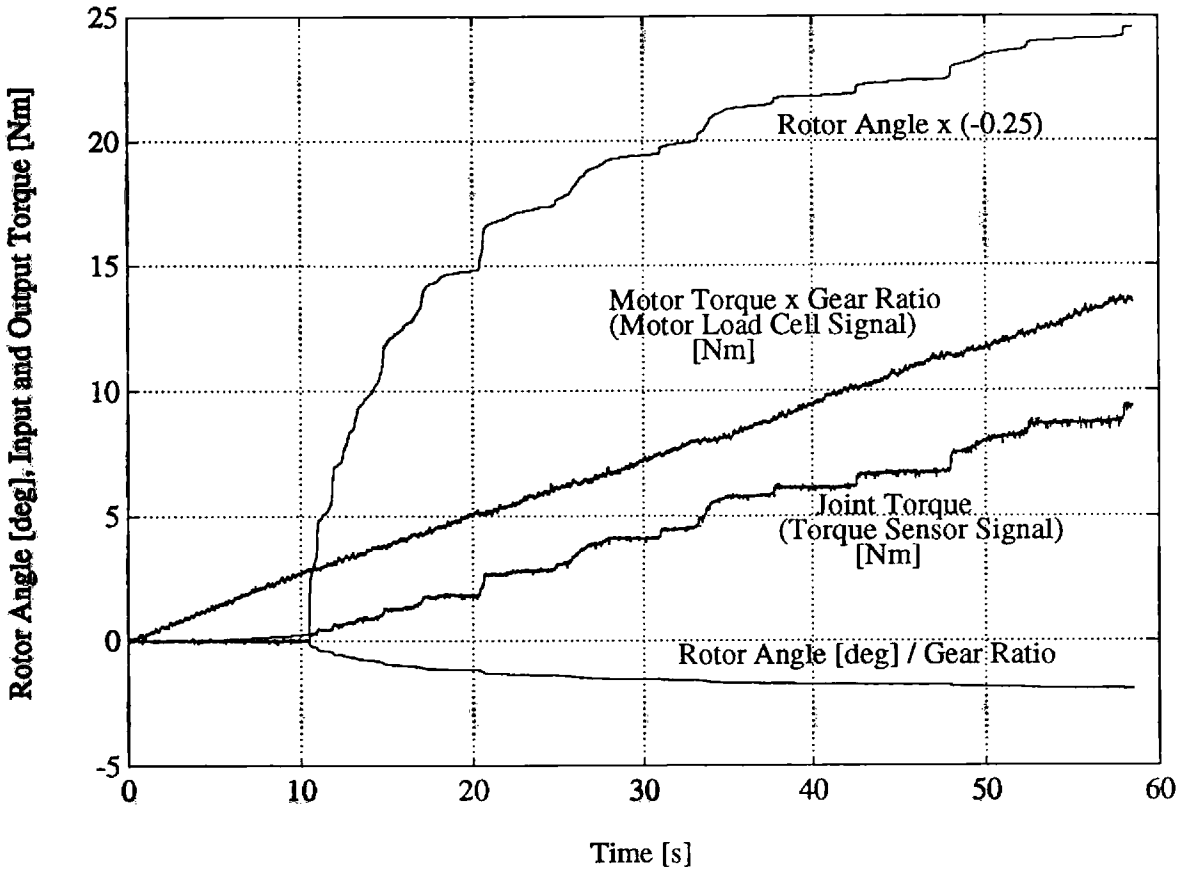


Fig. 8. Soft-contact experiment ("slow" ramp): a 5 mm sponge layer on a stiff surface.

sharp pulses of high magnitudes, due to elastic coupling between high-frequency "internal" and low-frequency "external" dynamics.

In contrast to dangerous instability when unrestrained, the behavior of the joint when the link is in full contact with a stiff environment is completely predictable under sinusoidal input, regardless of the frequency and amplitude of excitation. The reason for this is that the "external" stiff contact prevents dynamic interaction between the "fast" and "slow" subsystems, i.e., the internal fast rotating, and external slow rotating components. The joint torque was measured for a sinusoidal motor current with a sweeping frequency ranging from 0 to 300 Hz. Unfortunately, in the range above 100 Hz, the measured torque was too small, and the characteristics could not be reliably obtained. The Bode plot of the output/input torque-transfer function (Figure 14) clearly shows that a resonant peak appears at about 25 Hz, and that the slope of the magnitude characteristics is 40 dB/decade. The phase gradually changes toward $-180°$. Apparently, the joint behaves as a second-order system. Despite low amplitudes at higher frequencies (above 100 Hz), the occurrence of vibrations was confirmed by the sound perceptible over the whole frequency range.

3.4. Step Response—Unrestrained Motion

To examine the behavior of an IRIS joint module under step-current commands when the output link of the joint is unrestrained, we repeated the experiments of Tuttle and Seering (Tuttle 1992; Tuttle and Seering 1993), expecting to see unpredictable jumps in velocity signals. The time-response plots of position and velocity are given in Figure 15. The current command steps cover the full range of motor current, from 0 to 3 amps, as shown in Figure 16. We noticed that the 3-amp current level could only be reached at low velocities, because back emf limits the current to 2.6 amps when the velocity exceeds 250 degrees per second. Except for this nonlinearity, the velocity response was regular. Some small velocity fluctuation, due to harmonic-drive elasticity causing internal vibrations, was observed.

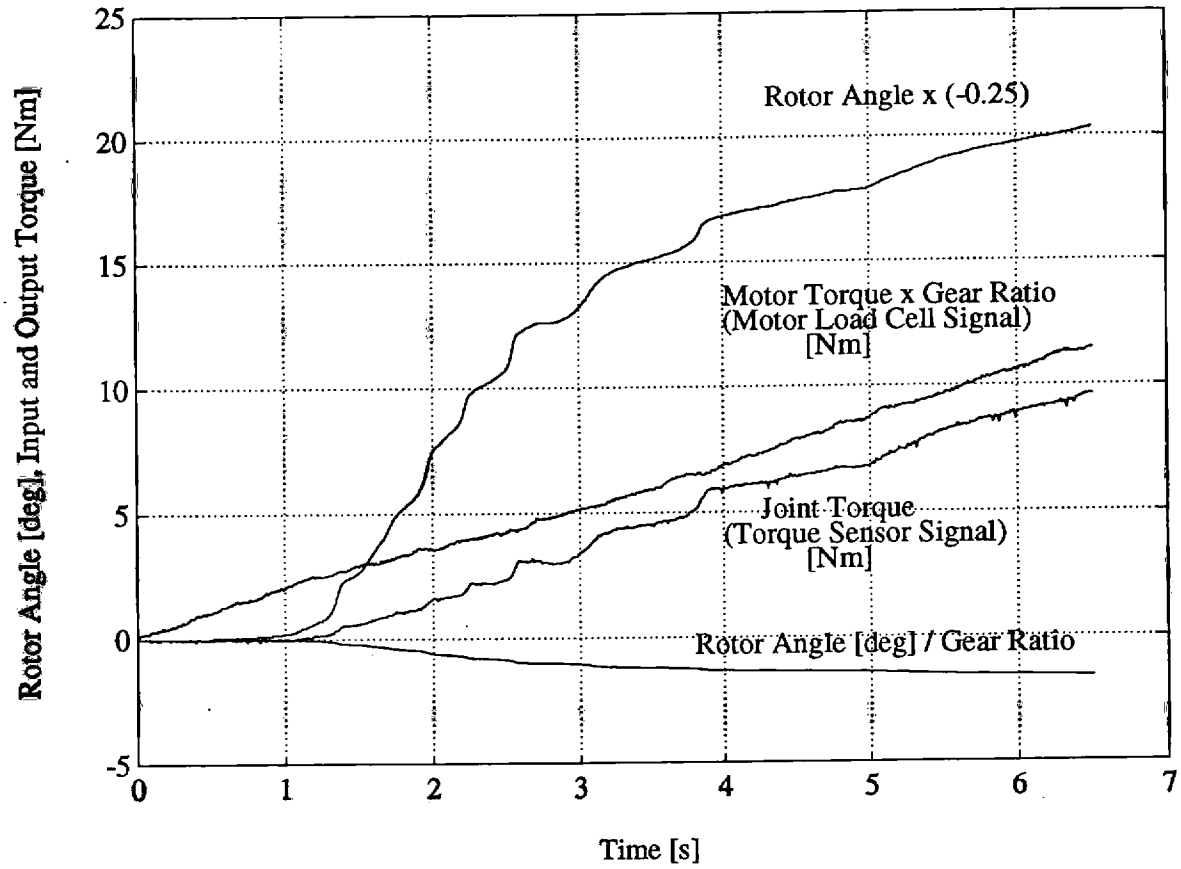


Fig. 9. Soft contact experiment ("fast" ramp): a 5 mm sponge layer on a stiff surface.

4. Mathematical Model

Better performance of a harmonic drive-based system can be achieved if one compensates for the effects of intrinsic nonlinearities and vibrations. To proceed in this direction, a general model of minimum complexity that allows control engineers to develop closed-loop controllers is developed in this section.

Equivalent mechanical and electrical diagrams that reflect basic joint properties are also included. Experimental observations described in the previous section are used for model verification.

4.1. Model Equations

The basic components of the IRIS joint (see Figure 3) are a DC motor, a harmonic drive gear, and a torque sensor. The entire system can be divided into four subsystems: motor amplifier, wave generator, flexspline, and circular spline. The last three subsystems are coupled through the harmonic drive. The description and mathematical formulation of each subsystem is given below.

4.1.1. Motor-Amplifier Subsystem

The frameless DC motor in partnership with the current-mode amplifier can be modeled as a "torque source" with a limiting capability, due to the action of the back emf and the current limiter of the amplifier.

The relationship between the control output u, motor current i_m, and rotor velocity $\dot{q}_w$ can be established easily as

$$i_m = k_s u, \qquad |i_m| \leq \min(I_s, (U_s - K_b|\dot{q}_w|)/r_m) \quad (1)$$

$$\tau_m = K_t i_m \quad (2)$$

where k_s is the amplifier gain, I_s is the upper bound of the current defined by the amplifier electronics, U_s is the maximum output voltage of the amplifier, K_b is the back-emf constant of the motor, r_m is the motor resistance, and K_t is the motor-torque constant. Notice that the nonlinearity observed in the motor-current response in Figure 16 is completely matched by eq. (1).

4.1.2. Wave-Generator Subsystem

The DC motor applies a torque τ_m to the input inertia, which is composed of the motor armature and the

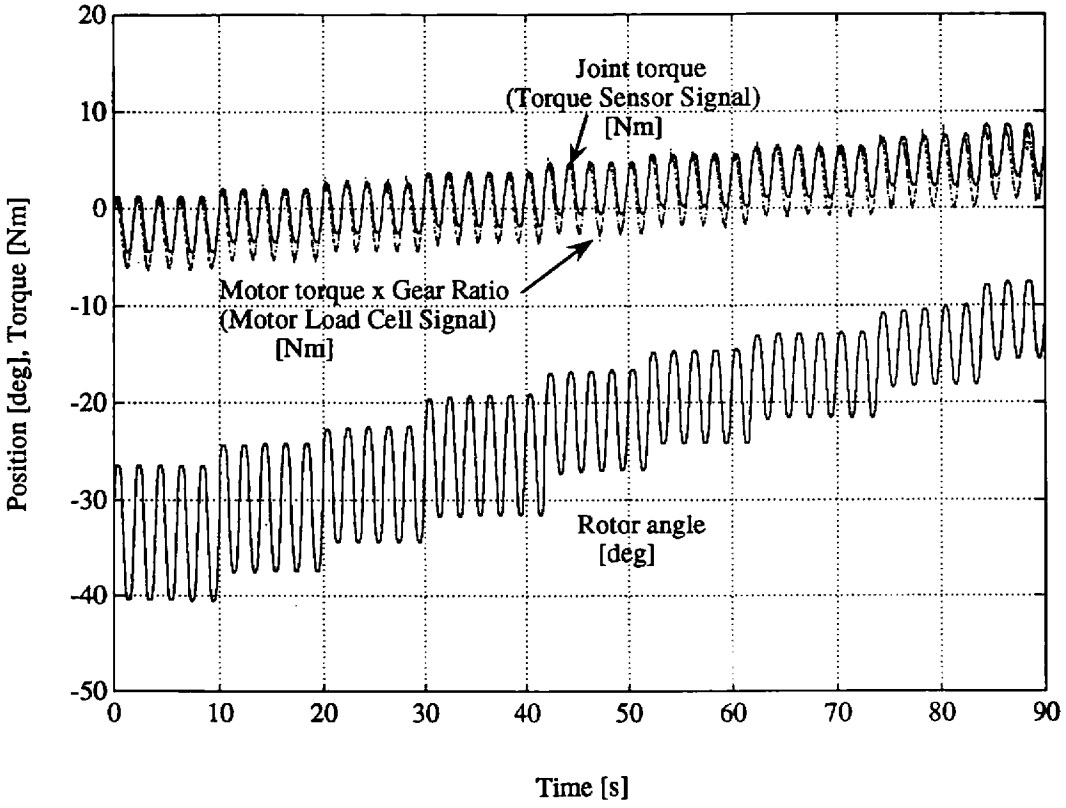


Fig. 10. Stiffness test.

harmonic drive wave generator. The torque driving the wave generator (which is ellipsoid in shape and surrounded by a ball bearing) is considered as the subsystem output. The following model involves Coulomb and viscous friction at the bearings **B1** and **B2**, and at the wave-generator ball bearing **Bwg** (see Figure 3):

$$\tau_m = J_{mw}\ddot{q}_w + \tilde{B}_m(\dot{q}_w) + \tilde{B}_w(\dot{q}_w + \dot{q}_f) + \tau_w \quad (3)$$

where $\dot{q}_f$ is the flexspline angular velocity with the reference direction opposite to the wave-generator velocity, J_{mw} is the combined inertia of the rotor, shaft, and the wave-generator, $\tilde{B}_w(\dot{q}_w)$ is the total friction torque generated at bearings **B1** and **B2**, $\tilde{B}_w(\dot{q}_w + \dot{q}_f)$ is the friction term of the wave-generator bearing **Bwg**, and τ_w is the torque exerted on the wave generator.

The friction term $\tilde{B}(\dot{q})$ involves kinetic (Coulomb), viscous, and static friction. The static friction includes Dahl and Stribeck terms that can be modeled as exponential or higher-order polynomial functions in $\dot{q}$ (Armstrong-Helouvry 1991). Rigorously, $\tilde{B}(\dot{q})$ is not a function, but a nonlinear hysteresis-type operator that depends on microscopic surface-contact forces. Unfortunately, the measurement of all these effects is highly complicated, especially if the gear is coupled with the motor and torque sensor and is enclosed within the joint frame. On the other hand, these effects usually appear only at very low velocities. For this reason, we will accept the model

$$\tilde{B}(\dot{q}) = B\dot{q} + (B^+, B^-)sgn(\dot{q}) \quad (4)$$

where B is a viscous coefficient, and B^+ and B^- are positive-direction and negative-direction Coulomb friction torques. The last term in eq. (4) is equal to B^+ for $\dot{q} > 0$, and is equal to B^- for $\dot{q} < 0$. This expression is consistent with the experimental results reported in the previous section (as shown in Figure 13).

If the flexspline is fixed to ground (input link), the velocity $\dot{q}_f$ is negligible compared to $\dot{q}_w$. If the flexspline is free to move, $\dot{q}_f = \dot{q}_w/N$ is still much lower than $\dot{q}_w$ since the typical value of the gear ratio N is 50, 100, etc. Taking this into account with the analysis above, the wave-generator subsystem model becomes:

$$\tau_m = J_{mw}\ddot{q}_w + \tilde{B}_{mw}(\dot{q}_w) + \tau_w \quad (5)$$

$$\tau_m = J_{mw}\ddot{q}_w + B_{mw}(\dot{q}_w) + (B^+_{mw}, B^-_{mw})sgn(\dot{q}_w) + \tau_w \quad (6)$$

We can include here a model of the torque-transmission characteristics described in Section 3.1. The torque exerted on the wave generator, τ_w, is not fully transmitted to the flexspline, due to the nonlinearity in the meshing

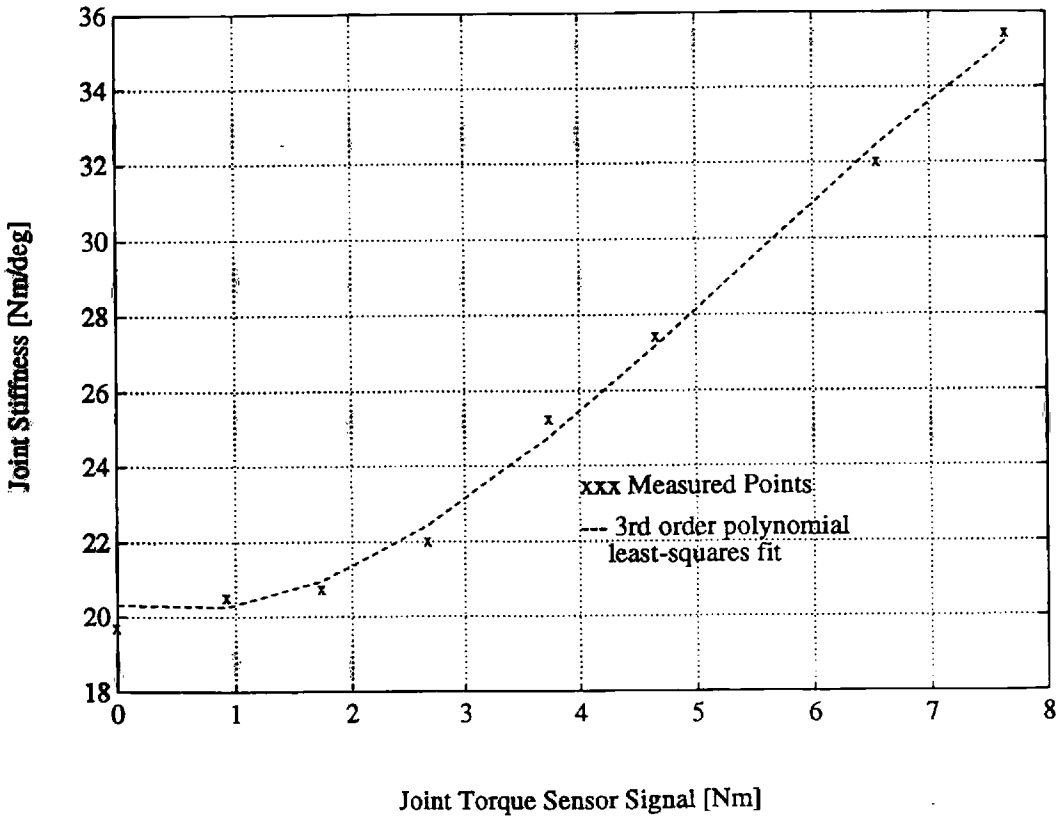


Fig. 11. Joint stiffness versus torque.

process. By neglecting the stick-slip component of the meshing process, but accounting for the torque-limiting feature, the simplest model would be

$$\tau_w = \tau_{ws} + \tau_{wnm} \tag{7}$$

where τ_{ws} stands for the component of the wave-generator torque that is fully transmitted to the output, while τ_{wnm} stands for the torque component due to irregular meshing of the flexspline-circular spline teeth. In other words, the torque τ_{ws} is of the form

$$\tau_{ws} = sat(\tau_w, \tau_{wc}(\dot{q}_w)) \tag{8}$$

where $sat(x, a)$ is the saturation function defined as $sat(x, a) = x$ for $-a \leq x \leq a$; otherwise, it is equal to $sgn(x)$. The critical torque τ_{wc} depends upon the velocity $\dot{q}_w$: If the velocity is low (as it is in "hard" contact tasks), τ_{wc} is significantly lower than the motor torque. In "soft" contact tasks or "free" motion, τ_{wc} is close to the motor torque.

The limited value of the wave-generator torque restrains both the flexspline and the circular spline torques to rise above corresponding limits. Thus, neither the internal torque sensor (mounted on the flexspline shaft) nor the external load cell (facing the output link attached to the circular spline) can measure the torques above the limit values.

A slightly simplified model of this phenomenon can be introduced by assigning the torque-limiting action to the motor torque τ_m. Then, a modified form of eq. (2) can be introduced:

$$\tau_m = sat(K_t i_m, \tau_{cr}) \tag{9}$$

where τ_{cr} is the "critical" torque at the input side that depends on $\dot{q}_w$ as described above.

4.1.3. Flexspline Subsystem

Experimental observations reveal three physical phenomena related to flexspline behavior: nonlinear stiffness, hysteresis, and quasi-backlash due to the soft-windup effect. We found that the following simple model adequately reflects the real behavior of the subsystem (see Figures 12 and 13):

$$\tau_f = \tilde{k}_f(q) + \tilde{B}_f(\dot{q}) \tag{10}$$

where τ_f is the torque transmitted to the flexspline, q is the flexspline torsion, and $\tilde{B}_f$ is the friction operator defined analogously to that given in eq. (4). The term

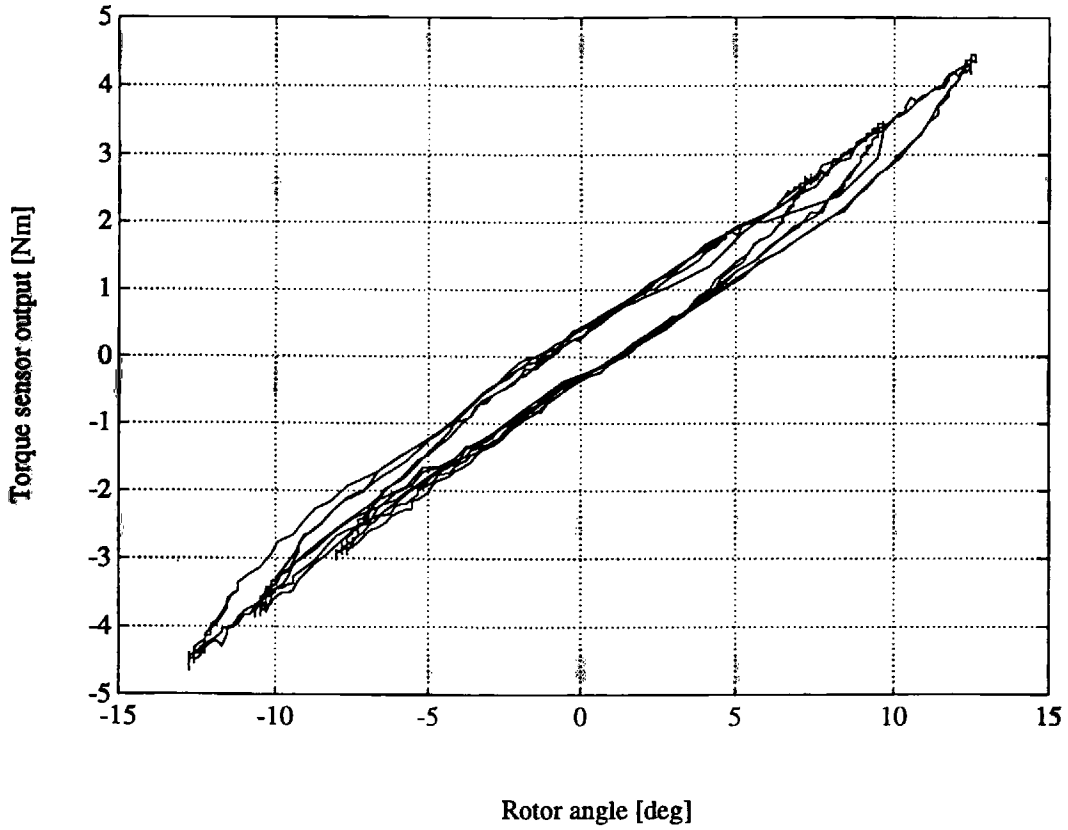


Fig. 12. Hysteresis effect.

$\tilde{B}_f(\dot{q})$ is introduced to match the hysteresis curve shown in Figure 12.

A careful analysis of the experimental data summarized in Figure 11 reveals that the stiffness coefficient of the harmonic drive, $k(q)$, can be modeled as a second-order polynomial in torsion q, except for the region close to zero, where the soft windup effectively decreases the joint stiffness to (see Figure 17):

$$k(q) = a_{f1}q^2 + a_{f2} + K_{sw}(q) \tag{11}$$

where a_{f1} and a_{f2} are coefficients, and $K_{sw}(q)$ is the soft-windup correction factor that can be modeled as a saddle-shaped function:

$$K_{sw}(q) = -k_{sw}e^{-a_{sw}q^2} \tag{12}$$

with k_{sw} and a_{sw} being the parameters that can be determined from the experimental plot in Figure 11. Consequently, the first term on the right side of eq. (10) can be presented as

$$\tilde{k}_f(q) = a_{f1}q^3 + a_{f2}q + K_{sw}(q)q \tag{13}$$

The torsion, q, is defined as $q = q_f - q_\tau$, where q_f is the angular displacement of the flexspline at the tooth engagement cross-section, while q_τ is the angular displacement of the torque sensor attached to the flexspline base (see Figure 3). The compact form of the flexspline model is thus

$$\tau_f = \tilde{k}_f(q_f - q_\tau) + \tilde{B}_f(\dot{q}_f - \dot{q}_\tau) \tag{14}$$

The torque sensor acts as a pure spring, with the stiffness coefficient k_τ much larger than that of the harmonic drive. The torque τ_f is measured by the torque sensor, i.e.,

$$\tau_f = \tau = k_\tau(q_\tau - q_F) \tag{15}$$

where q_F is the angular displacement of the F-end port of the joint (see Figure 3). This angular displacement is zero if the F end is grounded. The stiffness coefficient k_τ is usually determined using special calibration equipment before integrating the sensor with other joint components. The calibration setup consists of a custom-designed mechanical setup and a high-precision torque sensor.

4.1.4. Circular-Spline Subsystem

The output link to the joint can be attached to either the flexspline or the circular spline. In the first case, the circular spline port is grounded; in the second case, the

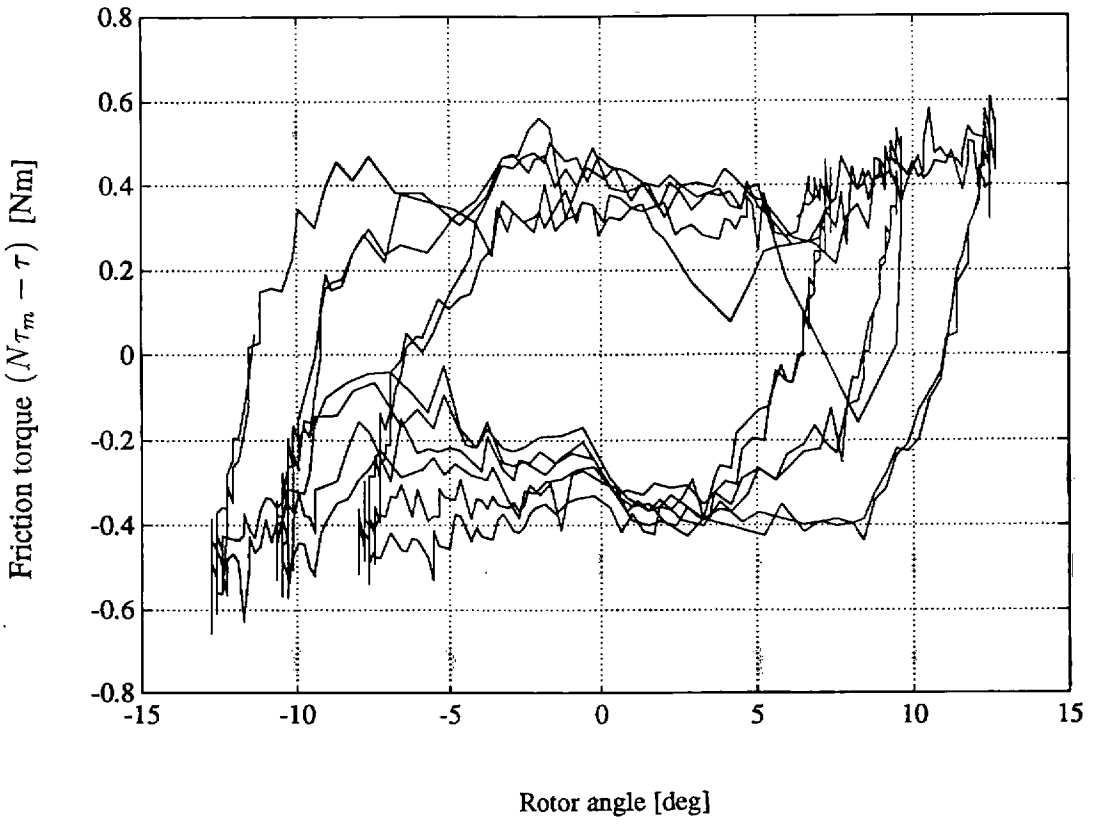


Fig. 13. Coulomb friction.

flexspline is connected to the input link. Therefore, one of the following two equations is used to describe C-end/F-end link dynamics:

$$\tau_c = J_c\ddot{q}_c + \tilde{B}_c(\dot{q}_c) + \tau_{Cl} \tag{16}$$

$$\tau = J_F\ddot{q}_F + \tilde{B}_F(\dot{q}_F) + \tau_{Fl} \tag{17}$$

where q_c is the C-end angular displacement, τ_{Cl} is the C-end load torque, and τ_c is the driving torque transmitted to the circular spline at the engagement zone cross-section. The parameters J_c and those in $\tilde{B}_c$ are inertia and friction parameters at the C end. The variables and parameters on the right side of eq. (17) are defined analogously.

4.1.5. Harmonic Drive

Having developed the dynamic equations of all subsystems, the entire system model can be integrated, provided that the harmonic-drive constitutive relationships are known. As explained earlier, the flexspline has two fewer teeth than the circular spline, and thus each full turn of the wave generator moves the flexspline two teeth in the opposite direction relative to the circular spline. This is a kinematic constraint given by

$$q_w = Nq_f + (N+1)q_c \tag{18}$$

where N is the gear ratio defined as $N = N_t/2$, and N_t is the number of teeth on the flexspline outer circumference. It is noteworthy that only one kinematic constraint relates the coordinates at the three ports. For this reason, one of the two output ports is often fixed to the ground. A second important observation is that the gear error is not represented by eq. 18. One way to introduce this error is given in Tuttle (1992); Tuttle and Seering (1993). The gear error can be electronically cancelled by properly modifying the motor current at the lowest control level as in Nye and Kraml (1991). For this reason, we shall not focus on this topic.

Applying the power-conservation law on the three-port device described by the kinematic constraint in eq. (18), or more precisely, on its derivative form:

$$\dot{q}_w = N\dot{q}_f + (N+1)\dot{q}_c \tag{19}$$

yields the input/output torque relationships (Tuttle 1992; Tuttle and Seering 1993)

$$\tau_f = N\tau_w \tag{20}$$

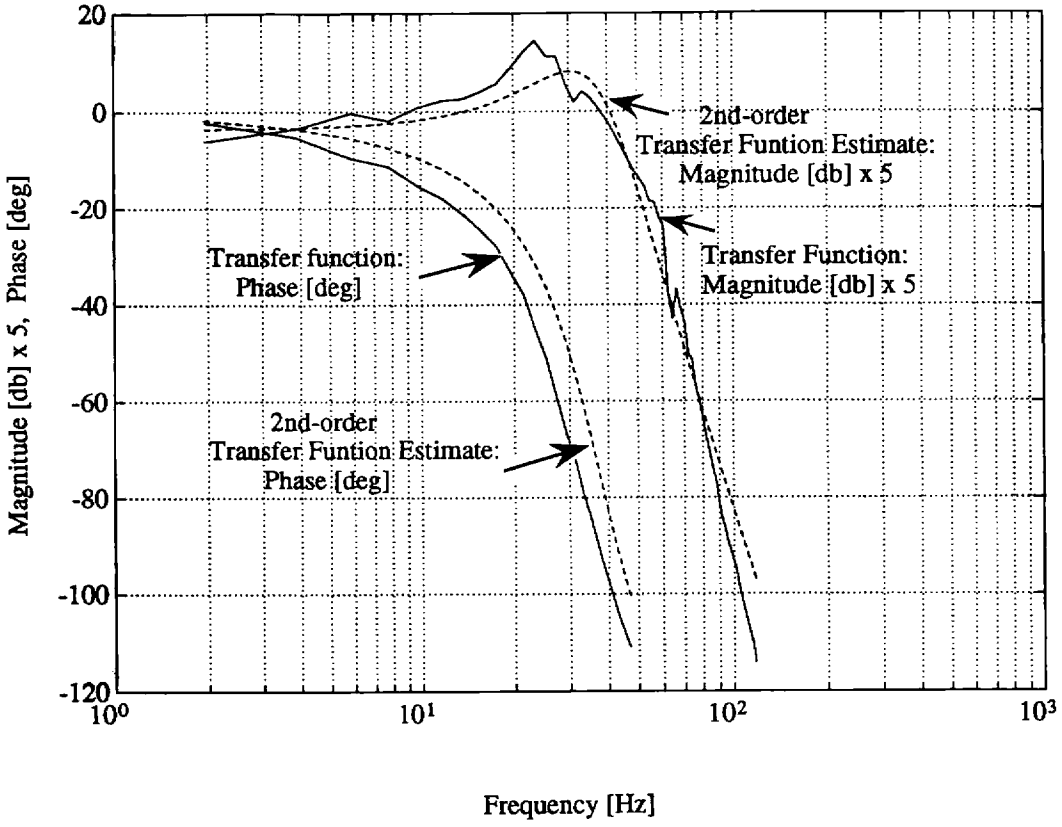


Fig. 14. Bode plot of input-output torque-transfer function.

$$\tau_c = (N+1)\tau_w \tag{21}$$

For building an intuitive understanding of the harmonic-drive operation, it is necessary to consider tooth-level geometry and interaction forces (Oei 1991; Tuttle 1992; Tuttle and Seering 1993). We have studied the distribution of forces interacting between the gear teeth, and established a framework for understanding the torque-transmission relations without introducing simplifications in the system. Here, we shall provide an explanation of the force-transmission mechanism, which can be easily formalized mathematically (Kircanski 1993).

A close look at the functioning of harmonic drive elements is provided in Figure 2. We used a magnifying glass to precisely observe the gear-tooth-meshing geometry during a set of our experiments. In contrast to the close-ups of the tooth-meshing geometry shown in Figure 3.24 in Tuttle (1992), we found the following. Since the meshing in the top and bottom halves of the harmonic drive (divided by the minor axis of the wave-generator ellipse) is symmetric, just the top half needs to be examined (see Figure 2). When looking directly at the center of the harmonic drive, it can be seen that the meshing is once again symmetric and mirrored between the left and the right halves. Thus, in the top-right half, the *right face* of each contacting gear tooth on the flexspline is in contact with the *left face* of each contacting gear tooth on the circular spline. The determination of the geometry of meshing is critical in analyzing the torque relationships.

To understand the torque-transmission property, suppose that the circular spline is fixed to ground, and that the flexspline is loaded by a rotational spring. The clockwise rotation of the wave generator (Figure 18) creates a load torque on the wave generator that opposes the driving torque, τ_w. The existence of the load torque can be explained as follows. The circular spline reaction force f_i, which is normal to the tooth i, acts on the flexspline tooth being engaged with the tooth i. Due to the elliptic shape of the wave generator, the force f_i has a normal component to the ellipse that creates the torque component $f_i \cos(\phi - \mu_i)\rho_i$, which acts against the driving torque, τ_w. Here, ϕ is the gear-tooth profile angle, μ_i is the angle between the normal to the circular spline and the normal to the flexspline at the contact point (tooth i in Figure 18), and ρ_i is the distance from the center of rotation to the normal to the ellipse at the same point. Taking into account all teeth being engaged, and integrating all torque components of the reaction torque, it can be shown that relationship (20) holds (Kircanski 1993).

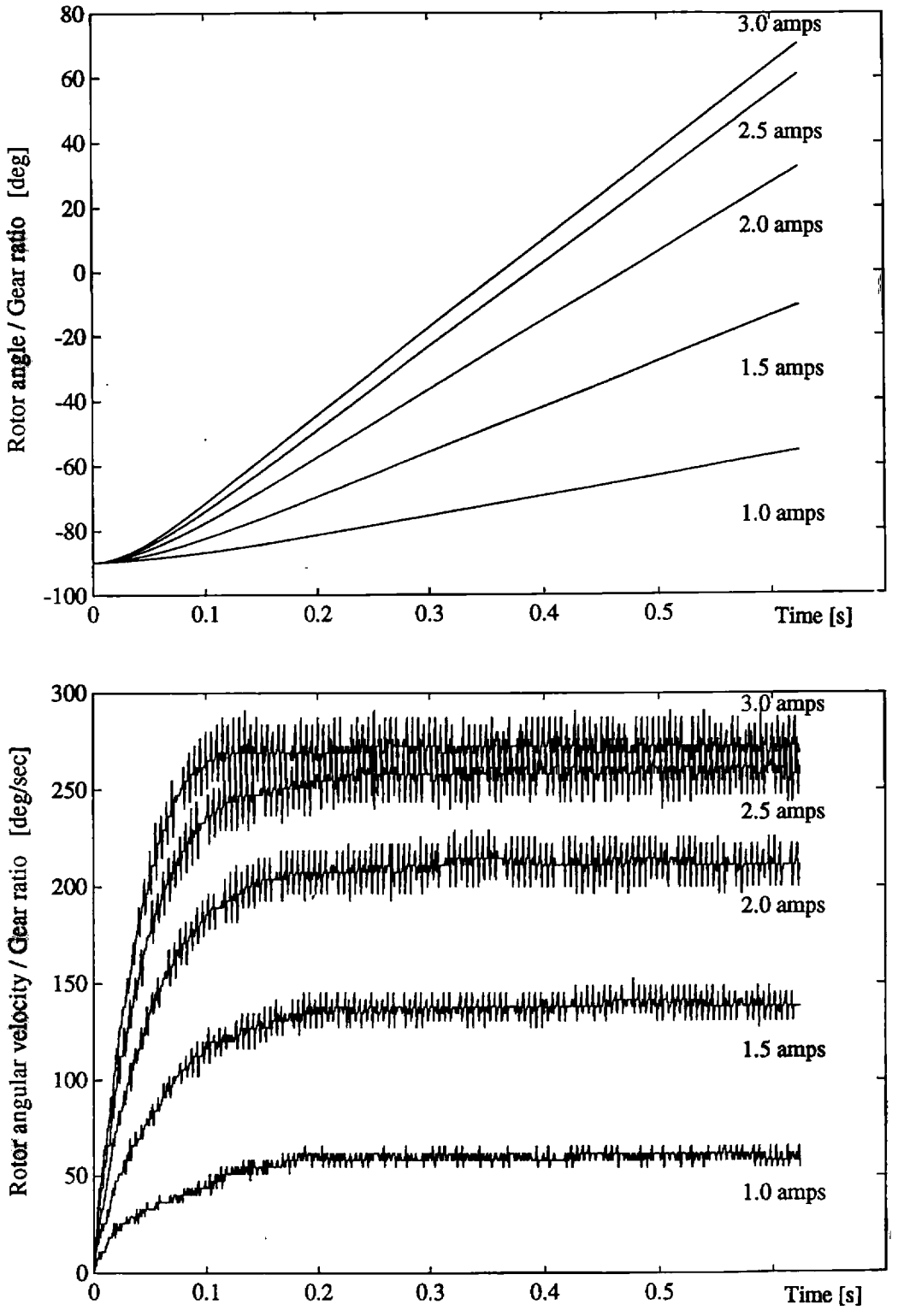


Fig. 15. Position and velocity step responses.

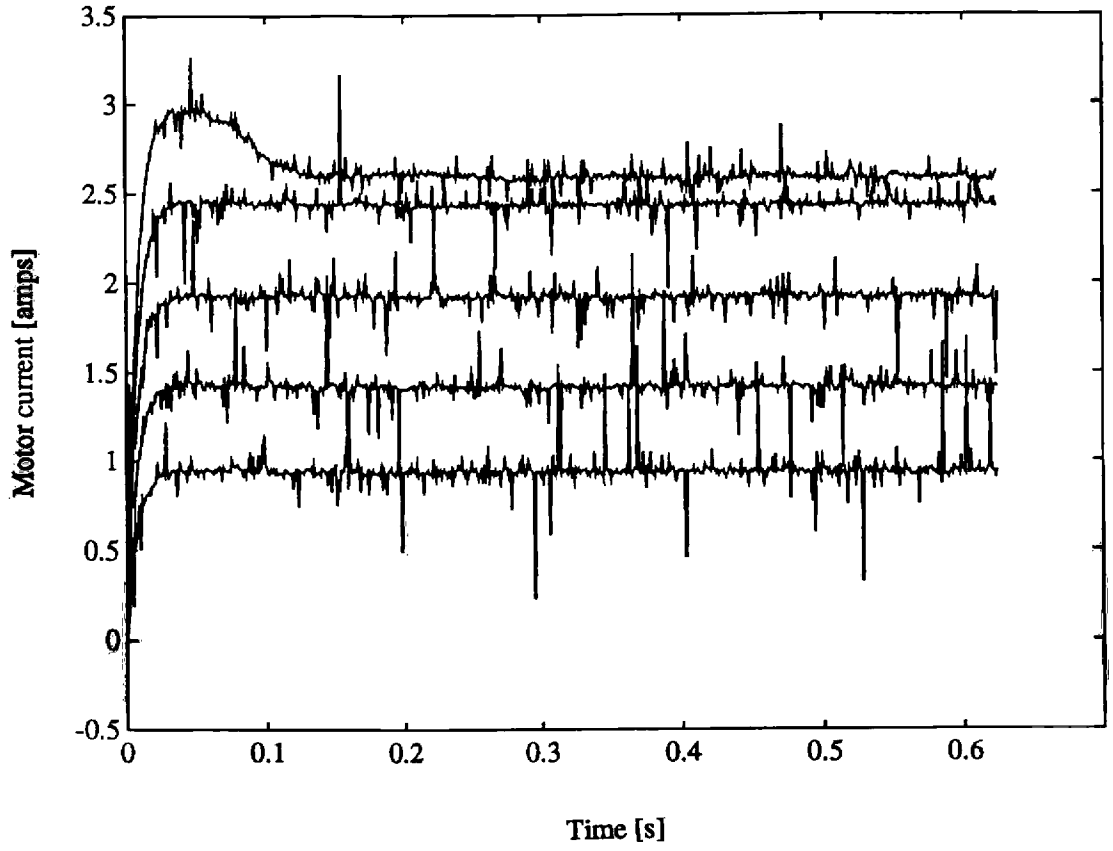


Fig. 16. Measured motor current.

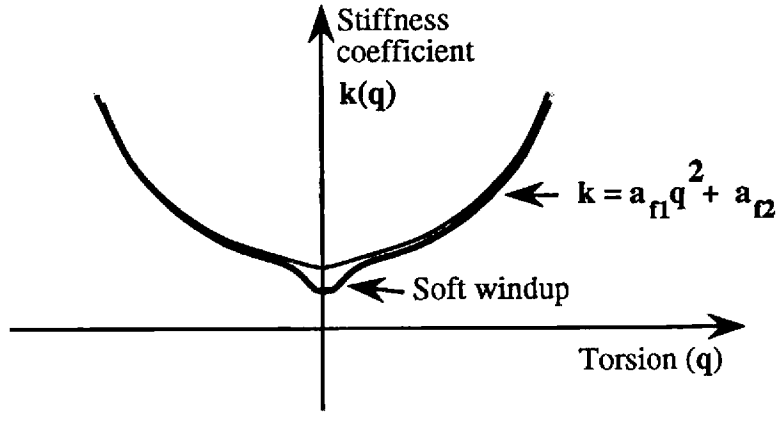


Fig. 17. Stiffness coefficient.

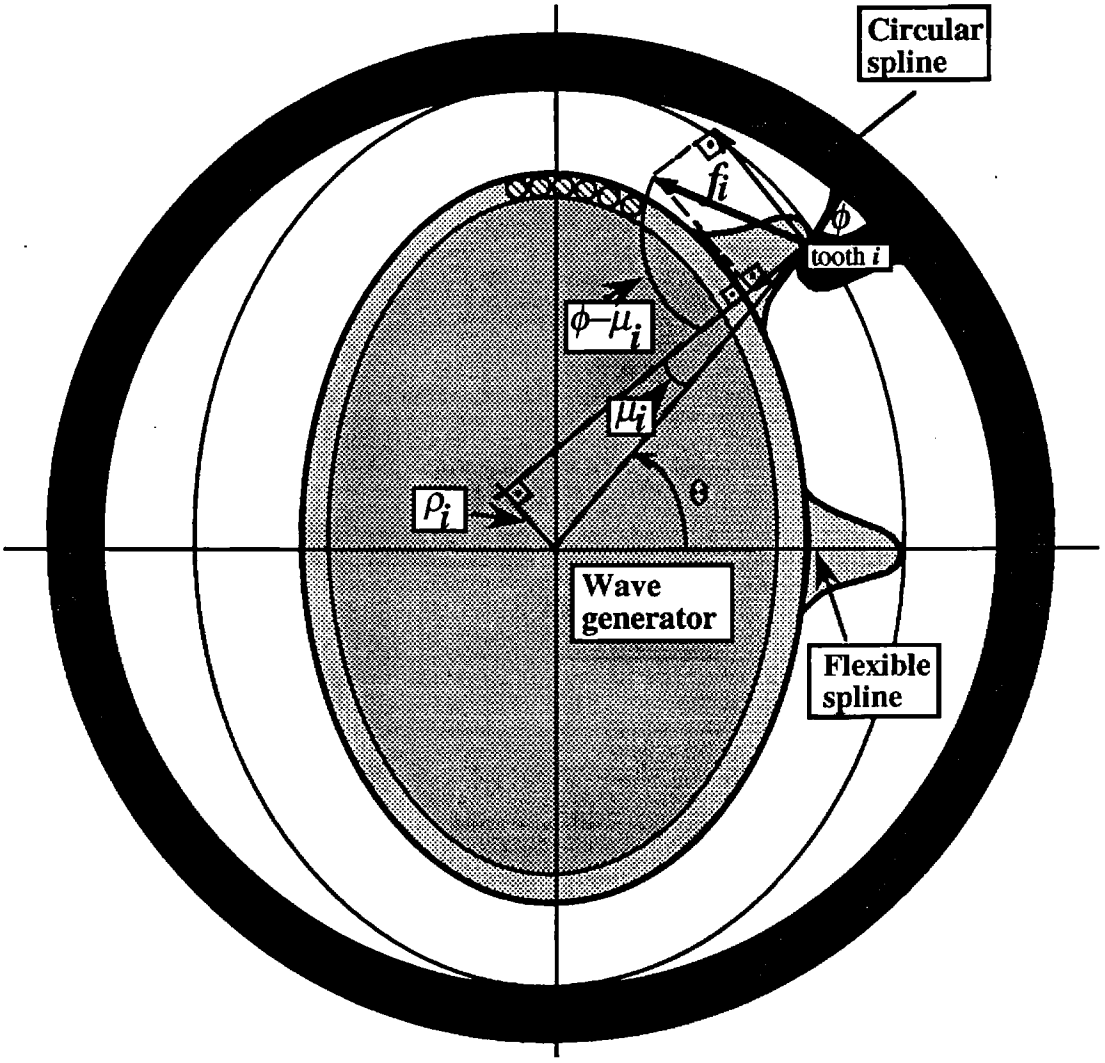


Fig. 18. Force distribution.

Since the flexspline has two fewer teeth than the circular spline, there exists a small phase shift between the corresponding teeth in engagement. Due to this phase difference, the sliding effect is imminent, causing energy losses and heating of the unit up to an equilibrium temperature of about 60–70°C. Clearly, any deviation of the teeth from an ideal shape results in a torque-transmission ripple, as well as the well-known errors in gear ratio. Another major source of gear error and torque-transmission ripple is out-of-roundness of the circular spline (Nye and Kraml 1991).

4.2. Equivalent Mechanical and Electrical Schemes

The models introduced in the previous section can be integrated into a single model that describes the kinematic and dynamic behavior of the robot joint. The following subsystem models have been introduced:

- PWM current-mode amplifier, eq. (1),
- DC motor, eq. (9),
- wave-generator subsystem, eqs. (5) and (6),
- flexspline subsystem, eqs. (14), (15), and (17),
- circular spline subsystem, eq. (16), and
- harmonic drive constraints, eqs. (18), (19), (20), and (21).

All of these equations constitute the joint model. An illustrative presentation of these equations is given in Figure 19 which represents an equivalent mechanical diagram of the joint. Notice that the harmonic drive is represented by an ideal three-port device, while all physical phenomena such as nonlinear stiffness, hysteresis, and friction

effects are extracted from the harmonic drive and attributed to components outside this idealized transmission device.

The mechanical diagram in Figure 19 can now be easily transformed into an equivalent electrical diagram, Figure 20. Note that there is a full analogy between a harmonic drive and a voltage transformer with two secondary windings. Both are three-port devices, and both exhibit hysteresis and have internal losses. Furthermore, both can be represented by ideal substitutes, while all losses can be attributed to external elements. Similarly, the Coulomb friction can be represented by a couple of Zener diodes, while the nonlinear spring corresponds to a nonlinear capacitor.

In the diagrams shown in Figures 19 and 20 the F and C ends are independently loaded by external loads. In practice, either the F or C end is grounded. This simplifies both equations and equivalent schemes. For example, suppose that the F end is fixed to the input link. In this case, in addition to the PWM amplifier and DC motor equations (1) and (9), the model reduces to the following three equations:

$$\tau_m = J_{mw}\ddot{q}_w + \tilde{B}_{mw}(\dot{q}_w) + \frac{1}{N}\tau \tag{22}$$

$$\tau = \frac{N}{N+1} g_c(\ddot{q}_c, \dot{q}_c, q_c, x) \tag{23}$$

$$\tau = \tilde{k}_f(\frac{1}{N}q_w - N'q_c) + \tilde{B}_f(\frac{1}{N}\dot{q}_w - N'\dot{q}_c) \tag{24}$$

where $g_c(\ddot{q}_c, \dot{q}_c, q_c, x)$ stands for C-end dynamics, and $N' = (N+1)/N$. The torque τ is the torque measured by the torque sensor. This torque is equal to the flexspline torque τ_f, as in eq. (15). These equations are the basis for the parameter-estimation procedure described in the next section.

5. Parameter Estimation and Model Evaluation

It is well known that estimating a system's unknown physical parameters, based on 1) a presumed system model, and 2) a set of experimental data, is the most reliable way to obtain accurate information about the model's validity and the parameter values. However, even if the model perfectly matches the system behavior, the estimates may be biased by the influence of sensor noise. For this reason, the model output must be compared with the real-system output under conditions that are different from those used for the parameter estimation. We applied two methods to eliminate bias, one based upon a "bias-compensating" least-squares algorithm (Zhao, Sagara, and Wada 1992; Jia 1993) and another based on an instrumental variables method (Jia 1993; Ljung 1987). The second method proved to be more reliable, especially when a noisy torque-sensor signal was used for parameter estimation.

The experiments described in Section 3 indicate that the dynamic behavior of the joint is highly dependent upon operating conditions. A physical phenomenon that dominates under one set of conditions may be literally nonexistent if the operating point is changed. For example, the flexspline hysteresis was a dominant effect only when the joint was stuck against an obstacle; otherwise, it was much less noticeable. For this reason, instead of estimating all parameters at once, it is much more convenient to divide the estimation process into three independent procedures corresponding to the four subsystems mentioned in Section 4. However, the parameters of the servo amplifier can be obtained by measuring the motor current and rotor velocity under different control outputs.

5.1. Motor Wave-Generator Parameters

By observing the model of the wave-generator subsystem, described by eqs. (5) and (6), it is clear that the Coulomb friction parameters B^+_{mw} and B^-_{mw} can be obtained from the experimental plot in Figure 13. In this experiment, the motor current was varied very slowly while the output link was kept fixed. The variations of q_w due to elasticity were small, so that the magnitudes of $\ddot{q}_w$ and $\dot{q}_w$ were negligible. In this case, the model of eq. (5) reduces to

$$\tau_m - (1/N)\tau = (B^+_{mw}, B^-_{mw})sgn(\dot{q}_w) \tag{25}$$

implying that the torque $\tau_m - (1/N)\tau$ is equal to the Coulomb friction torque. Since both τ_m and τ are measurable, the unknown constants on the right side of eq. (25) can be accurately estimated. From the experimental data (in Figure 13), we found $B^+_{mw} = 0.008$ N · m, and $B^-_{mw} = -0.007$ N · m.

As a second step, we introduced a composite sinusoidal excitation $u(t)$ that uniformly covered the joint bandwidth up to 5 Hz, and observed the unrestrained joint motion. We used the model of eq. (5) in the form

$$\tau_e = J_{mw}\ddot{q}_w + B_{mw}(\dot{q}_w) \tag{26}$$

to estimate the unknown inertia and viscous friction. The torque $\tau_e = \tau_m - (1/N)\tau - (B^+_{mw}, B^-_{mw})sgn(\dot{q}_w)$ was computed on-line from the motor current, joint-torque sensor signal, and motor velocity. We further multiplied both sides of eq. (26) with the gear ratio N, and replaced q_w with Nq_{wo}, where q_{wo} was the wave-generator angle as seen from the output port. So, we obtained

$$\tau_{eo} = (N^2 J_{mw})\ddot{q}_{wo} + (NB_{mw})\dot{q}_{wo} \tag{27}$$

with $\tau_{eo} = N\tau_e$. Note that the inertia and viscous friction terms ($N^2 J_{mw}$ and NB_{mw}) are no longer "small"

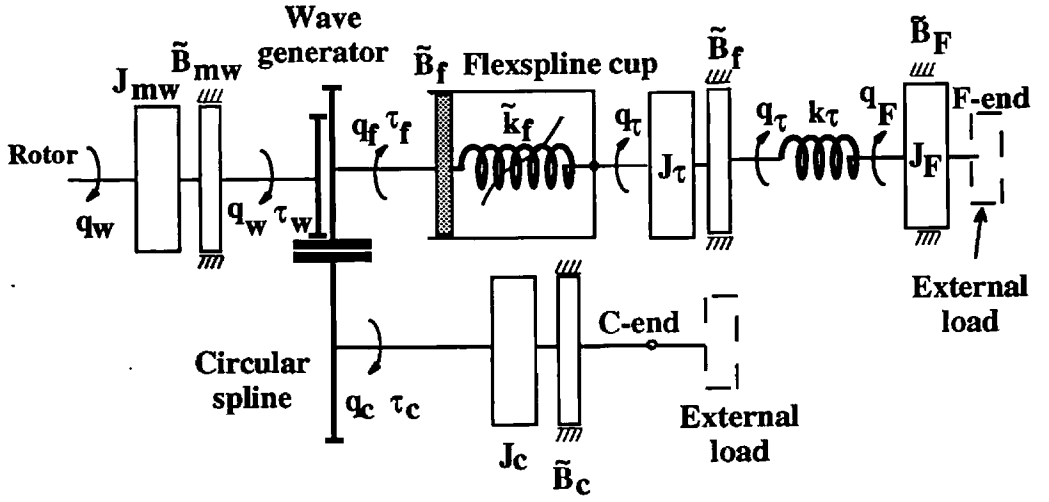


Fig. 19. Equivalent mechanical scheme.

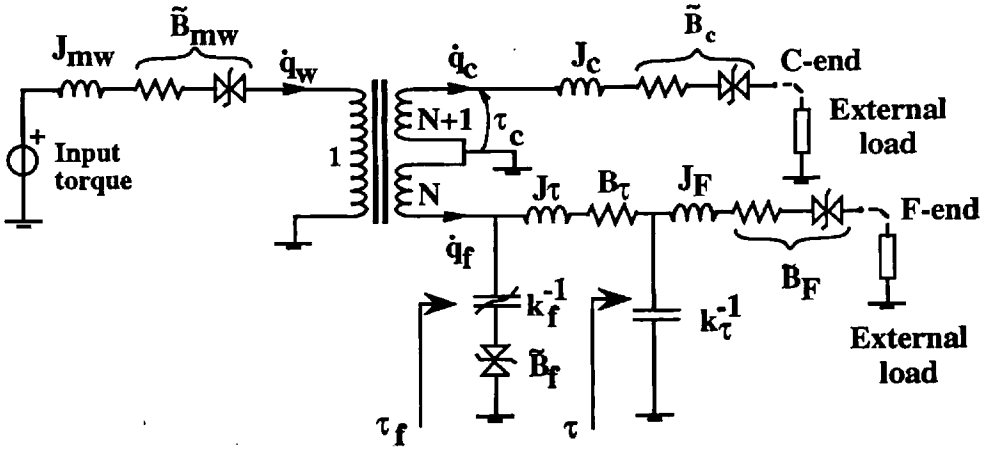


Fig. 20. Equivalent electrical scheme.

parameters that are therefore difficult to estimate. We then applied the bias-compensating least-squares method and obtained the following estimates: $J_{mw} = 8.0\,10^{-6}$ kgm^2, and $B_{mw} = 6.4\,10^{-4}$ N · m/(rad/s). Obviously, the system has significant damping in contrast to relatively low inertia.

To validate the assumption on the model structure and also the estimated parameters, we replaced the original trajectory with a modified one and performed the same experiment once again. Comparing the real and simulated responses (Figure 21) shows that the estimates are very well tuned to the real parameters.

To verify this conclusion, the same experiment was conducted with different payloads. This revealed that the wave-generator subsystem parameters are independent from the end-point mass. Experimental results with a 1-kg payload are shown in Figure 22. Notice that the torque-sensor signal is much larger than that in Figure 21. In this case, the estimated damping coefficient was $B_{mw} = 6.1\,10^{-4}$ N · m/(rad/s), which is about 5% lower than that with a zero payload.

5.2. Output-Link Parameters

The estimation of the output-link parameters is tremendously simplified by the use of the joint-torque sensor. The joint-torque measurement makes an overall system decomposition possible: eqs. (23) and (24) can be regarded as virtually independent, due to availability of $\tau(t)$.

Similar to the previous case, we used a composite sinusoidal excitation to cover the joint bandwidth as uniformly as possible. If the output link is moving in the horizontal plane, the model of eq. (23) becomes simple:

$$\tau = \frac{N}{N+1}(J_c\ddot{q}_c + B_c\dot{q}_c) \tag{28}$$

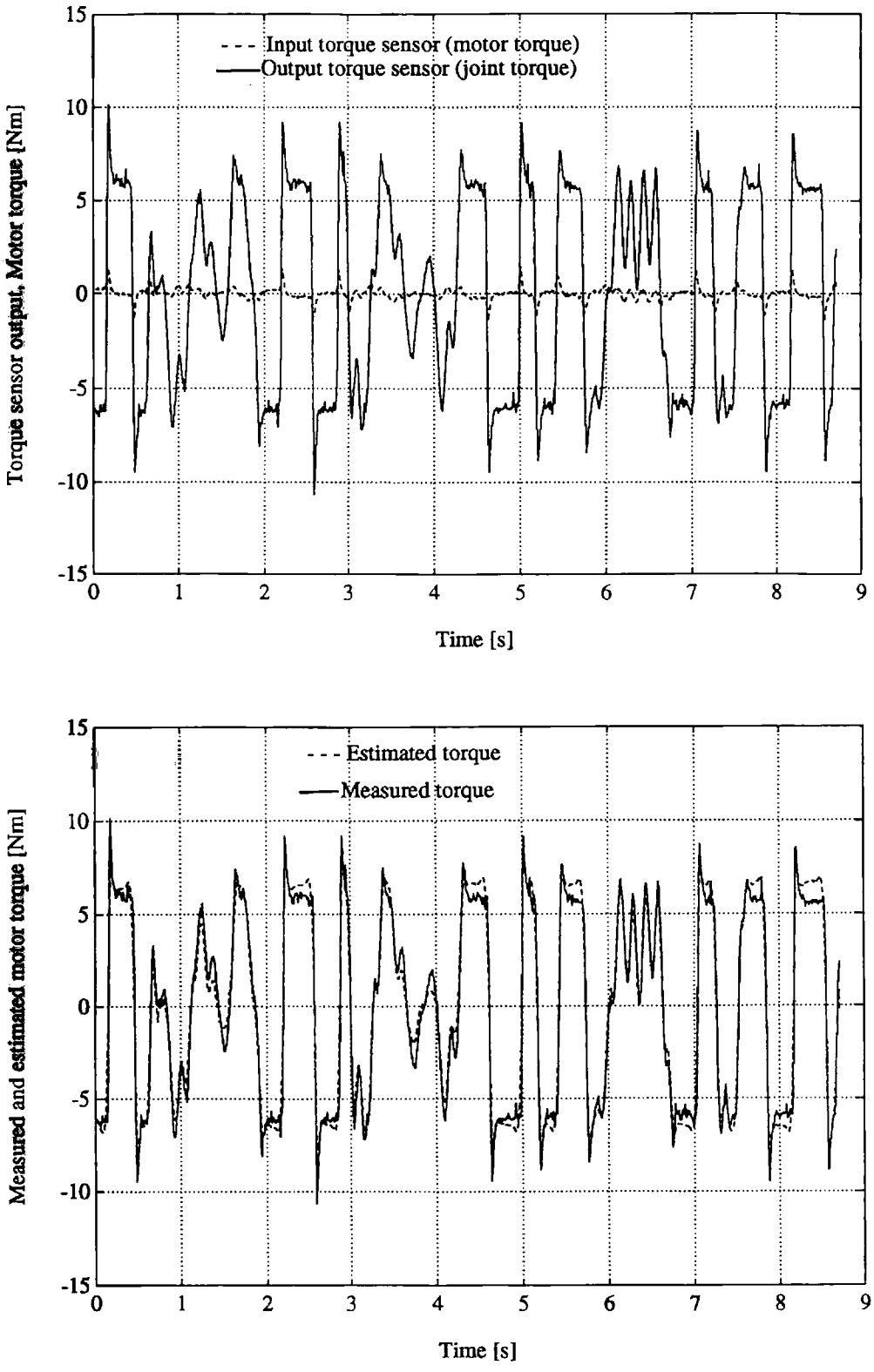


Fig. 21. Motor torque prediction based on estimated parameters for a 0-kg payload.

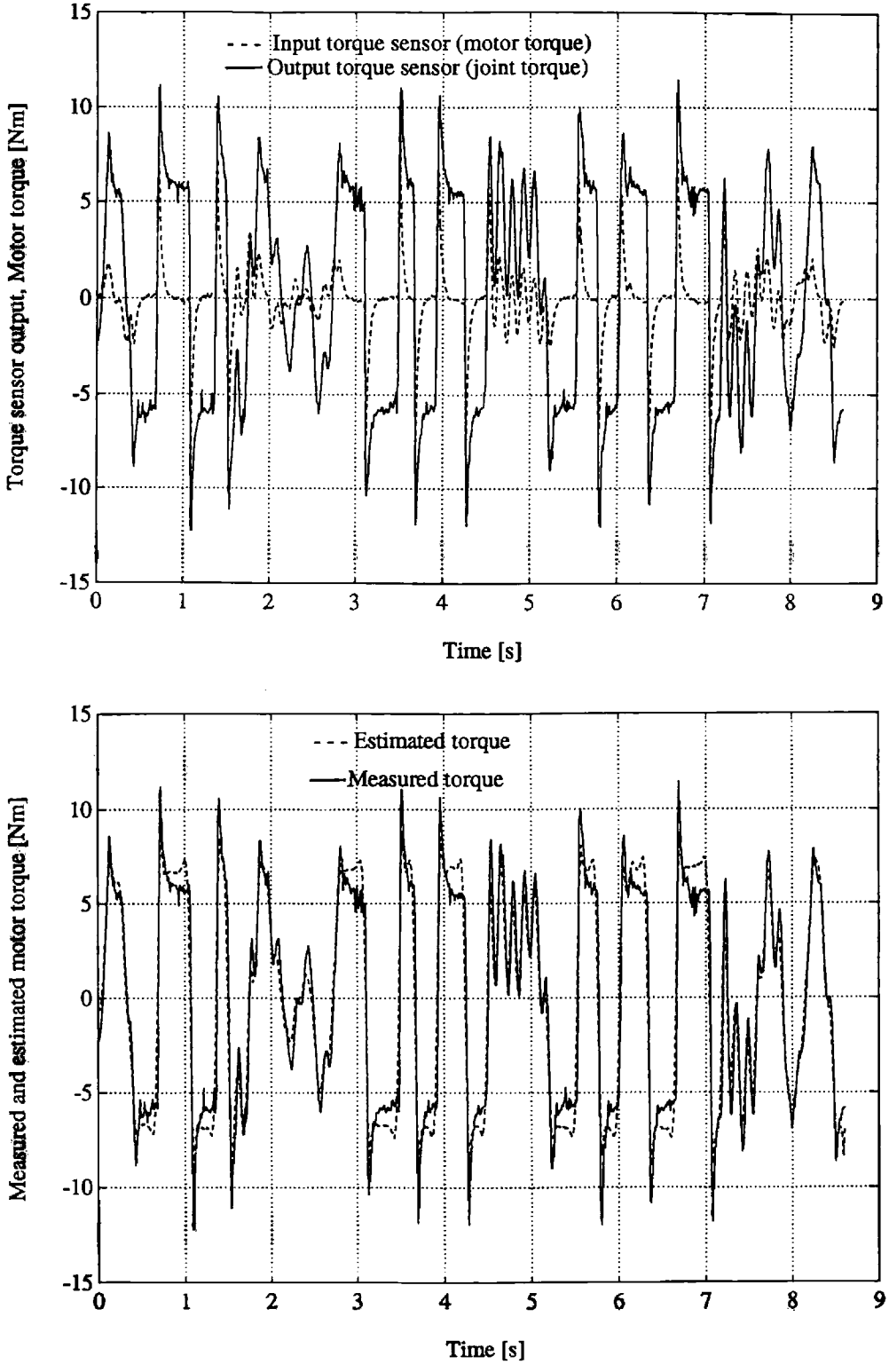


Fig. 22. Motor torque prediction based on estimated parameters for a 1-kg payload.

The unknown parameters estimated as $J_c = 0.013\,\text{kgm}^2$ and $B_c \approx 0$ were verified by comparing the measured and estimated torque-sensor outputs. These parameters correspond to a zero payload. In contrast to the wave-generator estimation results, the viscous friction was now negligible in comparison to inertia. It was shown that the viscous friction in the ball-bearing assembly between the elliptic body of the wave generator and the flexspline (**Bwg** in Figure 3) has much more influence on the wave-generator subsystem behavior than the output bearings (**B3** and **B4** in Figure 3) have on the C-end subsystem.

The second experiment with a 1-kg payload mounted at the end of a 0.24-m link was conducted to obtain a new set of parameters. The estimated inertia was $J_c = 0.06\,\text{kgm}^2$. We concluded that the actual added inertia ($1\text{kg} \cdot (0.24^2)\,\text{m}^2 = 0.057\,\text{kgm}^2$) is very close to our estimate.

5.3. Flexspline Parameters

To distinguish the flexspline subsystem behavior from both the input (wave-generator) subsystem and the output (circular-spline) subsystem, the torque-sensor signal was used again. Note that the flexspline model of eq. (24) is a quasi-algebraic model that encompasses nonlinear stiffness and hysteresis. If the output link is constrained, $q_c = 0$, then eq. (24) can be approximated as follows:

$$\tau = a_{f1}q^3 + a_{f2}q - k_{sw}e^{-a_{sw}q^2}q + (B_f^+, B_f^-)sgn(\dot{q}) \quad (29)$$

where $q = (1/N)q_w$ represents the flexspline torsion. Given a direction of motion, $sgn()$ becomes either 1 or -1, and the torque-torsion relation becomes purely algebraic. The parameters a_{f1}, a_{f2}, k_{sw}, and a_{sw} can be obtained by the use of the stiffness diagram in Figure 11, while the parameters B_f^+ and B_f^- are easier to obtain from the hysteresis plot in Figure 12. For the IRIS joint, we obtained the following values in SI units: $a_{f1} = 6.61, a_{f2} = 5.27, k_{sw} = 7.8, a_{sw} = 25.0, B_f^+ = 0.4$, and $B_f^- = -0.4$. Similar results were obtained for the case when the C end was grounded.

6. Conclusions

Based on experimental and theoretical studies, we have developed a systematic way to capture and rationalize the behavior of a harmonic drive-based robot joint. In addition, this article reveals an interesting phenomenon related to an intrinsic joint-meshing process. It was shown that in hard-contact tasks the output torque may not be equal to the input torque multiplied by the gear ratio. We also found that restrained motion experiments were more convenient than free-joint experiments for examining joint phenomena. Simple models of soft windup, hysteresis, and friction were established and used as constituents of the entire joint model represented by three basic nonlinear differential equations describing high-frequency, low-frequency, and coupling dynamics of rotor wave generator, circular spline, and flexspline, respectively. The significance of the joint-torque sensor has been highlighted from the standpoint of measurability of joint behavior. Finally, a systematic parameter estimation procedure has been proposed not only to obtain joint parameters, but also to support the validity of the developed model. This model is currently being exploited to enhance the performance of torque-feedback control algorithms.

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